

Causal Machine Learning for Heterogeneous SSRI Treatment Effects in Swedish Registries: Target Trial Emulation with High-dimensional Iterative Causal Forests

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Abstract

Background: Selective serotonin reuptake inhibitors (SSRIs) are first-line pharmacological treatment for depression in young people. Yet, randomized trials have linked their use to elevated suicidal behavior in adolescents and young adults. Trials are typically underpowered for rare outcomes and exclude high-risk patients, so observational evidence in representative populations is needed to characterize the absolute risk and to identify subgroups in which the effect varies.

Objectives: We aimed (1) to estimate the average causal effect of SSRI initiation on suicidal behavior risk among individuals aged 6–24 years with a new depression diagnosis, and (2) to investigate whether these effects differed across patient subgroups using iterative causal forest (iCF) methods.

Methods: We emulated a target trial in Swedish national-register data covering individuals aged 6–24 with a first depression diagnosis (ICD-10 F32 / F33) between 1 July 2006 and 31 December 2019, after at least one year without antidepressant dispensation. SSRI initiators were defined as those dispensing an SSRI (ATC N06AB) within 28 days of diagnosis, and non-initiators were the complement. The intention-to-treat analysis used inverse-probability-weighted Kaplan–Meier estimation of the 12-week risk of suicidal behavior (ICD-10 X60–X84 and Y10–Y34) from the index date. Heterogeneous treatment effects were investigated using the iterative causal

forest (iCF) algorithm of Wang et al. and the high-dimensional variant (hdiCF), which augmented the curated covariate set with ordinal frequency-encoded features built from individual ICD-10 and ATC codes. iCF subgroups were validated with a stratified IPW analysis using the same propensity-score model as the main analysis.

Results: Of 88,427 patients followed for 12 weeks, SSRI initiation was estimated to increase the absolute risk of suicidal behavior by +0.97 percentage points (95% CI 0.78, 1.17). The increase was larger in ages 6–17 than in 18–24 and larger in females than in males. The results were robust to excluding inpatient index diagnoses and to a 14-day grace period. The iCF analyses identified sex as the single stable effect modifier. Neither deeper iCF trees nor hdiCF augmentation identified additional reproducible subgroups.

Conclusions: In this cohort of Swedish youth, SSRI initiation following a depression diagnosis was estimated to produce a small but consistent increase in 12-week suicidal-behavior risk, broadly aligned with prior target-trial-emulation evidence in the same age range.

Keywords: SSRI, depression, suicidal behavior, target trial emulation, causal inference, causal machine learning, heterogeneous treatment effects, iterative causal forest

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1 Introduction

1.1 Clinical background

Depression is among the most prevalent mental disorders worldwide [1], with a high and increasing burden among adolescents and young adults [2]. Selective serotonin reuptake inhibitors (SSRIs) are the recommended first-line pharmacological treatment for depression in both adults and young people [3]. Prescribing rates have increased substantially in recent decades: in Sweden, antidepressant dispensation among individuals aged 0–24 rose from 1.4% to 2.1% between 2006 and 2013 [4], and a Scandinavian comparison found that incident use nearly doubled in Sweden from 2007 to 2018 [5].

Despite this growing exposure, the safety of SSRIs in young populations remains contested. Since the early 2000s, there have been concerns about a potential association between SSRI use and increased suicidal ideation and behavior [6, 7]. A meta-analysis of randomized controlled trials reported an approximately twofold increased risk of suicidal ideation or attempts in children and adolescents treated with antidepressants [8], though the evidence in adults is more mixed. More recently, Lagerberg et al. [9] used Swedish national registers to emulate a target trial and estimated the causal effect of SSRI initiation on suicidal behavior within 12 weeks of a depression diagnosis. They found an increased risk among individuals under age 25, consistent with the RCT evidence.

Together, these findings motivate deeper investigation into how the treatment effect varies across subgroups of young people with depression. If certain patients are at disproportionate risk, identifying them could inform clinical decision-making. For example, clinicians might choose more intensive monitoring for high-risk subgroups, or choose psychotherapy over pharmacotherapy when the expected harm from SSRIs outweighs the benefit. The FDA black-box warning already stratifies risk by age (under 25) [10], but it does not differentiate by sex, clinical history, or other patient characteristics that may further modify the treatment effect.

1.2 Methodological background

Randomized trials of SSRI suicidal-behavior risk in young people have limitations. Suicide attempts and completed suicides are rare, so individual trials are typically underpowered for these outcomes [11]. Trial follow-up is usually short: weeks rather than months [9]. The clinical subgroups at highest baseline risk, such as individuals with prior suicidal ideation or behavior, are routinely excluded from trial enrollment [12, 13]. As an alternative to RCTs, observational studies of population-scale registry data can generate evidence on rare but serious outcomes in populations representative of those who receive antidepressants in clinical practice.

Using observational data to estimate causal effects exposes the analysis to confounding by indication, immortal time bias, and time-varying confounding. To address these, observational data can be analyzed within the framework of target trial emulation (TTE). TTE applies the design principles of randomized trials, explicitly defining the hypothetical trial that the analysis aims to emulate and then structuring the analysis to mirror that trial as closely as possible [14, 15]. While TTE does not remove bias from unmeasured confounding, it enables a structured approach that minimizes other common biases [16]: it resolves immortal time bias by aligning eligibility, treatment assignment, and time zero, and provides a transparent template for handling confounding by indication and time-varying confounding through inverse probability weighting.

Heterogeneous treatment effects (HTEs) describe how the effect of an intervention varies across patient subgroups [17]. Traditional subgroup analysis addresses HTE by stratifying on pre-specified variables and testing for treatment–covariate interactions. However, this approach has well-documented limitations: it requires pre-specifying which variables to examine, cannot discover complex multi-variable interactions, and is prone to inflated false-positive rates from multiple testing. With 10 subgroup tests at 5% significance, the family-wise error rate reaches approximately 40% [18]. Furthermore, interaction tests have roughly four times the variance of main-effect tests, so conventional subgroup analyses are often underpowered to detect real heterogeneity even when it exists [19].

Causal machine learning (CML) methods, such as causal forests [20, 21], offer a complementary approach. Rather than testing pre-specified hypotheses, causal forests search the full covariate space to discover subgroups with differential treatment effects in a data-driven manner. Each tree is fit using honest estimation: one subsample chooses the splits, while a disjoint subsample estimates the treatment effect within each leaf, preventing the same observations from being used to both define and confirm a subgroup [20]. A single cross-validated heterogeneity test then replaces the multiplicity of pre-specified subgroup tests, controlling overall Type I error [22]. The iterative causal forest (iCF) algorithm [23] further improves stability through plurality voting across many iterations, and the high-dimensional variant (hdiCF) [24] additionally incorporates automatically generated features from ICD-10 and ATC codes to probe for effect modifiers that may be missed by a curated covariate set. Applying these methods to SSRI safety in young people tests whether causal ML can refine the FDA’s age-only risk stratification with patient-level effect modifiers.

1.3 Objectives

This study addresses two complementary research questions:

- Substantive question: Do SSRIs increase the risk of suicidal behavior in certain

subgroups of youth (6–24 years) diagnosed with depression?

- Methodological question: Can high-dimensional iterative causal forests (hdiCF) be used to uncover clinically meaningful patient subgroups with heterogeneous treatment effects in Swedish registry data?

The primary objective is to estimate the causal effect of SSRI initiation following a depression diagnosis on the risk of suicidal behavior within 12 weeks among individuals aged 6–24 years. Secondary objectives include:

1. Applying iCF and hdiCF methods to identify patient subgroups with heterogeneous treatment effects
2. Evaluating the stability, interpretability, and clinical plausibility of the identified subgroups

2 Methods

2.1 Data sources

We used the following Swedish national registers, linked via pseudonymized personal identification numbers:

- National Patient Register (NPR): Inpatient (since 1987) and outpatient (since 2001) diagnoses
- Prescribed Drug Register (PDR): All dispensed prescriptions since July 2005
- Total Population Register: Demographics and migration
- Cause of Death Register: Date and cause of death
- LISA database: Socioeconomic variables (education, income)
- Multi-Generation Register: Parental linkages for family history

2.2 Study population

The study was conducted based on the target trial emulation framework [9]. Eligible diagnoses were recorded between 1 July 2006 and 31 December 2019 (both inclusive). Follow-up continued through the administrative end of study on 31 December 2020 (Section 2.5). Individuals were eligible if they:

1. Had a first recorded diagnosis of depression (ICD-10: F32 or F33, excluding F33.4) between July 2006 and December 2019
2. Were aged 6–24 years at diagnosis

3. Had no antidepressant (ATC: N06A) dispensation on any of the 364 days preceding the diagnosis date (dispensations on the diagnosis date itself are treated as the index treatment event, not as prior exposure)
4. Were registered in Sweden at the time of diagnosis

Figure 1 presents the cohort inclusion flowchart.

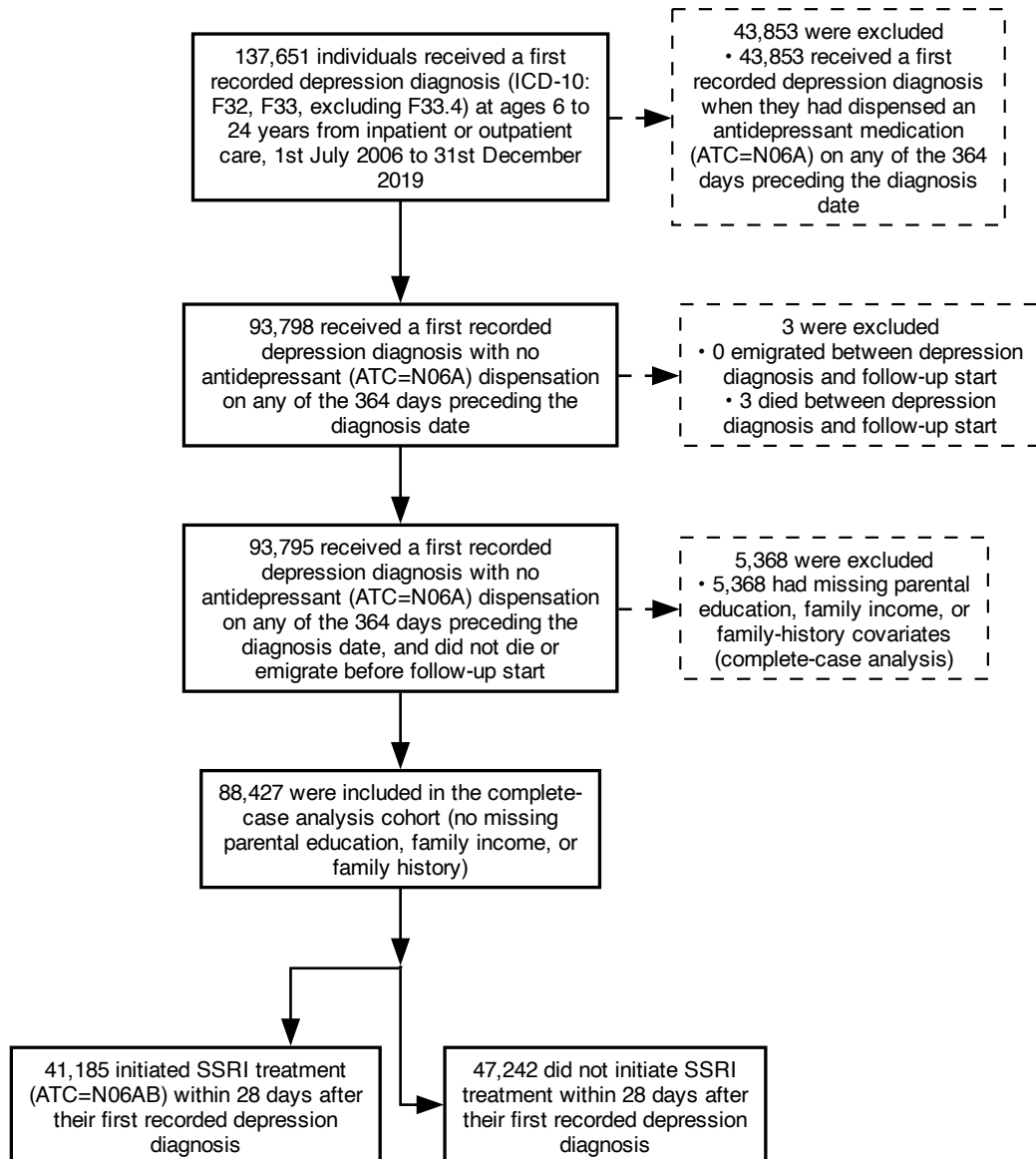


Figure 1: Flowchart of cohort inclusion criteria and study population derivation.

2.3 Treatment definition

SSRI initiators: Individuals who filled an SSRI prescription (ATC: N06AB) on the date of the depression diagnosis or any of the 28 days following it.

Non-initiators: Individuals who did not fill an SSRI prescription within 28 days.

The 28-day grace period allows for clinical assessment and treatment decisions following diagnosis [9]. The empirical distribution of the diagnosis-to-first-dispensation lag among initiators is shown in Figure 30 (Appendix B).

2.4 Outcome definition

The primary outcome was suicidal behavior, defined as:

- Intentional self-harm (ICD-10: X60–X84)
- Events of undetermined intent (ICD-10: Y10–Y34)

Events were identified from the NPR (inpatient and outpatient) and the Cause of Death Register.

2.5 Follow-up

Follow-up began at the date of SSRI dispensation for initiators. For non-initiators, follow-up start was assigned using a frequency-matching approach to ensure comparable time-zero distributions, as in Lagerberg et al. [9].

Follow-up ended at the earliest of:

- Suicidal behavior event
- Death
- Emigration
- 12 weeks (84 days) after follow-up start
- Administrative end of study (December 31, 2020)

Events occurring on the same day as follow-up start were not counted as outcomes. The SSRI dispensation was treated as pre-exposure on day 0, so any same-day suicidal-behavior event is interpreted as having preceded any pharmacological effect of the drug.

2.6 Covariates

Baseline covariates used for confounding adjustment are listed in Table 1. Diagnosis covariates were defined as any occurrence in the NPR on or before the index date (follow-

up start). Medication covariates were defined as any dispensation in the 90 days up to and including the index date. The mapping from the underlying register-code variable names in the extraction pipeline to the thesis-facing labels used throughout this report is given in Table 13 (Appendix A).

Table 1: Baseline covariates used for confounding adjustment

Variable	Description
Calendar year	Year of index diagnosis (used in nuisance models only)
Care setting	Source of the index depression diagnosis in the NPR. Coded as O = outpatient, S = inpatient (“sluten vård”), or T = other/unknown source. For modelling, O and T are merged into a single non-inpatient category contrasted against S.
<i>Demographics</i>	
Sex	0 = male, 1 = female
Age	Continuous; categorized into 0 = 6–11, 1 = 12–17, 2 = 18–24 years for iCF analyses
<i>Socioeconomic factors</i>	
Parental education	Highest parental education level (0 = primary, 1 = secondary, 2 = post-secondary, 99 = missing)
Family income	Family disposable income, categorized as 1 = negative, 2 = zero, 3 = 1st–20th percentile of positive income, 4 = 20th–80th percentile, 5 = >80th percentile, 99 = missing
<i>Family history (parental, ICD-10)</i>	
Suicidal behavior	X60–X84 (intentional self-harm), Y10–Y34 (undetermined intent); 0/1/2 = count of biological parents with at least one such code; 99 = missing parental linkage
Depression	F32–F33; 0/1/2 = count of biological parents with at least one such code; 99 = missing parental linkage
<i>Prior psychiatric diagnoses (ICD-10, binary)</i>	
Alcohol use disorder	F10
Substance use disorder	F11–F19, excluding tobacco (F17)

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(Table 1 continued)

Variable	Description
Psychotic disorder	F20–F29 (schizophrenia and other psychotic disorders)
Bipolar disorder	F30–F31
Phobic anxiety disorder	F40.0–F40.2 (agoraphobia, social phobias, specific phobias)
Other anxiety disorder	F41.0–F41.1 (panic disorder, generalized anxiety disorder)
Obsessive-compulsive disorder	F42
Stress/adjustment disorder	F43
Anorexia nervosa	F50.0–F50.1
Bulimia nervosa	F50.2–F50.3
Sleep disorder	F51 (non-organic sleep disorders)
Cluster B personality disorder	F60.2–F60.3 (dissocial and emotionally unstable personality disorders)
Intellectual disability	F70–F79
Autism spectrum disorder	F84.0/.1/.5/.8/.9
ADHD	F90 (hyperkinetic disorder)
Conduct disorder	F91
Overdose/poisoning	T36–T51, X40–X49
Prior suicidal behavior	X60–X84 (intentional self-harm), Y10–Y34 (undetermined intent)
<i>Prior medications (ATC, within 90 days before baseline, binary)</i>	
Antipsychotic	N05A, excluding lithium (N05AN)
Hypnotic/sedative	N05C
Benzodiazepine	N05BA
Antiepileptic	N03A, excluding mood stabilizers (N03AG01, N03AX09, N03AF01)
Mood stabilizer	N05AN (lithium), N03AG01 (valproate), N03AX09 (lamotrigine), N03AF01 (carbamazepine)
Psychostimulant	N06B
Opioid	N02A
Addiction medication	N07B (nicotine, alcohol, or opioid dependence treatments)
<i>Healthcare utilization (binary)</i>	

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(Table 1 continued)

Variable	Description
Prior psychiatric hospitalization	Any inpatient psychiatric admission before baseline

Four of the curated categorical covariates (parental education, family income, family history of suicidal behavior, and family history of depression) had non-trivial fractions of missing values arising from incomplete registry coverage or missing parental linkages in the registers. The patterns of missingness across these four variables are reported in Table 14 (Appendix A.2). We restricted the analysis cohort to complete cases for these four variables. In total, 5,368 patients (5.72% of the post-washout, post-pre-fu eligible cohort) were excluded due to missing at least one value. Complete-case restriction was applied uniformly to the main intention-to-treat (ITT) analyses, the sensitivity analyses, and the iCF / hdiCF subgroup analyses, so all downstream estimates share the same cohort. Family income was treated as a factor with five levels (negative, zero, 1st–20th percentile, 20th–80th percentile, >80th percentile) in the propensity-score model, following Lagerberg et al. [9]. For the iCF/hdiCF analyses it was retained on its integer scale so the forest could split between any pair of consecutive levels.

2.7 Statistical analysis

2.7.1 Intention-to-treat analysis

We used inverse probability weighting (IPW) to estimate the intention-to-treat effect. Stabilized weights were computed as:

$$SW_i = \frac{P(A = a_i)}{P(A = a_i | L_i)} \quad (1)$$

where A is treatment assignment, a_i is the observed treatment for individual i , and L_i is the vector of baseline covariates.

The treatment probability $P(A = 1 | L)$ was estimated using logistic regression. Weights were truncated at the 99th percentile to reduce the influence of extreme weights.

The primary estimand was the 12-week risk difference in suicidal behavior comparing SSRI initiation to no SSRI initiation. Cumulative incidence at 12 weeks was estimated as $1 - \hat{S}(t)$, where $\hat{S}(t)$ is the weighted Kaplan-Meier survival function. Risk differences and risk ratios with 95% confidence intervals were calculated.

2.7.2 Sensitivity analyses

Two pre-specified sensitivity analyses assess robustness of the main ITT estimate:

1. Exclusion of inpatient diagnoses: The main ITT analysis was repeated on the cohort restricted to individuals whose index depression diagnosis was made in a non-inpatient setting (outpatient or other/unknown setting). Inpatient diagnoses are a small fraction of the cohort but capture a clinically heterogeneous group with substantially different baseline suicide risk compared to outpatient-only patients [25], and the IPW model has more uncertainty for these patients. Comparing the restricted-cohort ITT estimate with the main estimate quantifies how much the main result depends on the inpatient subset.
2. Grace period: To assess sensitivity to the 28-day grace-period choice on treatment-group assignment, we rebuilt the cohort end-to-end under a stricter 14-day grace period. Patients whose first SSRI dispensation fell within 14 days of diagnosis are initiators, and the rest are non-initiators. The non-initiator follow-up start is frequency-matched to the new 14-day initiator distribution (rather than the 28-day one), and the death/emigration eligibility filter spans 14 days rather than 28.
3. Missing-indicator method: To assess how much of the complete-case restriction (Section 2.6) drives the headline result, we re-fit the ITT analysis on the full eligible cohort ($N = 93,795$) without dropping rows on the four partially-observed covariates. A single binary indicator (`any_miss`) marks patients with sentinel-99 in at least one of parental education, family income, family history of suicidal behavior, or family history of depression, and was added as a covariate in the PS model. For every patient with `any_miss = 1`, all four covariates were set to their modal reference level (parental education = secondary; family income = 20th–80th percentile; both family-history variables = 0), so the indicator behaves as a clean “missing-stratum” flag whose coefficient absorbs the entire contribution of those patients via the four affected covariates. All other modelling choices match the headline ITT analysis. Known biases of this method are discussed in Section 4.4.

Heterogeneity across age, sex, and additional covariates is addressed separately in the data-driven HTE analysis described in Section 2.8.

2.8 Heterogeneous treatment effect analysis

To identify patient subgroups with differential treatment effects, we applied the iterative causal forest (iCF) algorithm [23]. This method extends standard causal forests [20] by iteratively refining subgroup definitions to improve stability and interpretability.

2.8.1 Overview of iterative causal forests

Causal forests are an ensemble machine learning method that estimates heterogeneous treatment effects by partitioning the covariate space into regions with different conditional average treatment effects (CATEs). The iCF algorithm improves upon standard causal forests through:

1. Variable selection: An initial causal forest identifies variables with importance above the mean, reducing the feature space to the most relevant effect modifiers. Importance is the depth-weighted frequency with which each covariate is used as a split across the trees in the forest, with splits near the root counted more heavily than deeper splits.
2. Iterative tree construction: Multiple causal forests are grown at each candidate tree depth. The best tree from each forest (minimizing the R-loss objective of Nie and Wager [26], as used by iCF [23]) is extracted.
3. Plurality voting: Across iterations, the most frequently occurring tree structure is selected, ensuring stability of subgroup definitions.
4. Synthetic tree construction: Split values are averaged across trees with matching structure to define final subgroup boundaries.
5. Cross-validation: K-fold cross-validation is used to select the optimal tree depth, balancing model complexity and predictive performance.

2.8.2 Implementation

The iCF analysis was conducted with the following specifications:

- K-fold cross-validation: $K = 5$ folds
- Trees per forest: 200
- Iterations per depth: 50 (cross-validation), 1000 (final model)
- Candidate depths: 2, 3, 4, and 5
- P-value threshold for heterogeneity test: 0.1

Age was categorized into three clinically meaningful groups: children (6–11 years), adolescents (12–17 years), and young adults (18–24 years). Calendar year was included in the nuisance models (propensity score and outcome regression) for confounding adjustment but excluded from the causal forest splitting candidates, since year is a confounder (related to both treatment propensity and outcome trends) but not a plausible effect modifier.

The raw causal forest used for variable selection (step 1 of the iCF pipeline) is grown without `grf`'s built-in hyperparameter tuner (`tune.parameters = "none"`, with an explicit RNG seed). Diagnostics on this cohort found that the auto-tuner's choice of hyperparameters varied substantially across random seeds. The CV-MSE landscape that the tuner navigates is flat (relative range $\sim 10^{-4}$, well below within-fold noise), so the tuner picks essentially at random among configurations of roughly equivalent CV fit. With tuning enabled, the above-mean-VI variable-selection step retained between 4 and 14 variables across five seeds, with the most extreme draws producing aggressive-shrinkage forests in which most candidate covariates received zero splits (Appendix A.5). Disabling the auto-tuner and using `grf`'s honest defaults removes this source of nondeterminism without measurably degrading the downstream voted-tree results.

Prior to running iCF, a heterogeneity test was performed using the calibration test from the initial causal forest to assess whether there is evidence of treatment effect heterogeneity [20].

Subgroup-specific conditional average treatment effects (CATEs) were estimated using inverse probability of treatment weighting within each identified subgroup. Confidence intervals were obtained via bootstrap resampling.

2.8.3 Raw causal forest and individual-level predictions

The initial causal forest grown in step 1 of the iCF pipeline (prior to variable selection and iterative refinement) also produces individual-level CATE predictions $\hat{\tau}(X_i)$ for each patient. These predictions are useful for evaluating the overall degree of treatment effect heterogeneity through calibration analysis, Qini curves, and the area under the targeting operator characteristic curve (AUROC) [22]. This provides a complementary perspective to the iCF subgroup analysis. Whereas iCF produces discrete, interpretable subgroups, the raw causal forest's continuous CATE predictions capture the full spectrum of heterogeneity.

To assess robustness of the individual-level CATE predictions, we additionally compared three CATE estimators on the same cohort: the causal forest, a doubly-robust learner, and a T-learner. Calibration plots for all three are reported in Figure 6.

2.8.4 High-dimensional iCF (hdiCF)

To assess whether the curated covariate set might miss relevant effect modifiers buried in the full diagnosis history or prescription history, we also applied the high-dimensional iterative causal forest (hdiCF) algorithm of [24]. The hdiCF extends iCF by generating ordinal frequency-encoded features from individual ICD-10 and ATC codes recorded before baseline, allowing the algorithm to discover finer-grained effect modifiers (e.g., specific

diagnosis codes) that are not represented in the curated covariate set.

This design differs from the original hdiCF specification in Wang et al. [24], where HD features replaced rather than supplemented predefined covariates. Wang used 200 most-prevalent codes per data dimension with no curated covariate set. We retained the curated covariates because they encode clinical knowledge (aggregated diagnosis categories matched to known suicide risk factors, family-history flags, and class-level medication exposure) that purely code-based HD features would only approximate. The trade-off is candidate redundancy: a curated flag such as `diag_alcohol` and the HD features `dx.inp.F10` / `dx.out.F10` encode overlapping signal, which can split the iCF iterations' best-tree votes between correlated candidates and depress each one's plurality fraction. We accepted this redundancy on the assumption that the curated and HD variants are complementary in granularity: the curated set absorbs broad clinical categories while the HD set adds setting-specific resolution and ordinal code-frequency information that the curated flags collapse over — and that the iCF voting procedure is sufficiently robust to consolidate on whichever variant best captures effect modification. The empirical outcome on this cohort is reported in Section 3.5.10.

Three classes of high-dimensional (HD) features were generated:

- Inpatient diagnosis codes (`dx.inp.*`): ICD-10 3-character codes recorded in an inpatient setting on or before the index date
- Outpatient diagnosis codes (`dx.out.*`): ICD-10 3-character codes recorded in an outpatient setting on or before the index date
- Prescription codes (`rx.*`): ATC 5-character (chemical-subgroup) codes dispensed in the 90 days up to and including the index date, excluding all antidepressants (ATC N06A); aligned with the curated medication covariates

Antidepressants (N06A) are excluded from the HD prescription feature set to avoid treatment leakage. Since the exposure is itself an N06AB dispensation, an inclusive 90-day window would make any N06A feature a near-perfect proxy for treatment rather than a baseline effect modifier. The curated medication covariate set already excludes N06A, so this filter keeps the two covariate sets free of antidepressants.

Each HD feature was encoded ordinally using patient-level frequency quartiles per Wang 2025 (0 = no occurrence, 1 = once, 2 = sporadic [2 to 75th percentile], 3 = frequent [>75th percentile]). Codes with population prevalence below 1% were excluded, and up to 200 codes per data dimension were retained. The realized feature counts for this cohort are reported in Section 3.5.10; the full per-dimension breakdown of generated HD features is given in Table 16 (Appendix A).

Per Wang 2025 recommendations for hdiCF, two methodological modifications were ap-

plied relative to base iCF:

1. Heterogeneity test skipped: Sparse HD features dilute the calibration test signal, so the test is bypassed.
2. Top 10% variable selection: The above-mean VI threshold used in base iCF is too permissive when many sparse features are present; instead, only the top 10% of variables by importance are retained for growing the iCF.

All other iCF parameters (K-fold CV, tree depths, iteration counts, plurality voting) were held identical to the base iCF analysis.

2.9 Software

All analyses were conducted in R version 4.5.1. The `survival` package was used for Kaplan-Meier estimation, and custom code was developed for IPW calculations. Causal forests were implemented using the `grf` package [20]. The iterative causal forest algorithm was implemented following the methodology of [23]. Source code for the extraction and analysis pipelines is available at <https://github.com/Sebelino/ssri-suicidality>. Results reported in this thesis correspond to the `thesis-submission` Git tag. The full study can be reproduced end-to-end, including the compilation of this thesis report. Access to the Karolinska Institutet Tensor HPC cluster is a precondition for this.

2.10 Ethical considerations

The study was approved by the Swedish Ethical Review Authority (Dnr: 2020-06540). All analyses were performed on pseudonymized data delivered by Statistics Sweden and the National Board of Health and Welfare to Karolinska Institutet. Individual consent was waived under the Swedish ethical-review framework for register-based research, where contacting the underlying population would be both infeasible and disproportionate given that the data have already been collected for administrative purposes. Source data are held on Karolinska Institutet’s secure Tensor cluster, and no direct or indirect identifiers appear in this thesis or its supplementary tables.

The data are nevertheless sensitive in several respects. The cohort consists of children, adolescents, and young adults aged 6–24, a group whose capacity to consent is limited even when consent is sought. Records include psychiatric diagnoses, antidepressant dispensations, and suicidal-behavior events. Family-history covariates are derived from each biological parent’s own register records. Third-party health data has been linked without any opportunity for the parents to consent or object. The pseudonymization, the secure-storage requirement, and the absence of any individual-level publication or re-identification are the practical safeguards for this kind of register-based research.

Aside from data protection, there are a couple of ethical considerations worth flagging. The headline finding is that there exists a sex difference in early-period suicidal-behavior risk at SSRI initiation. This finding could in principle be misused to delay or withhold treatment from female patients on grounds of attributed harm. Section 4.5 frames the finding as an additional axis of caution to be raised in a risk-benefit discussion at SSRI initiation, not as a basis for differential prescribing. This distinction matters because untreated depression itself carries substantial morbidity and the absolute risk increase reported here is small (roughly one additional suicidal-behavior event per hundred patients treated over 12 weeks). The deeper iCF subgroups, which surface candidate modifiers based on parental education and family income, could similarly be misread as identifying “high-risk” subgroups based on socioeconomic status. We frame these explicitly as exploratory and not reproducible across iterations precisely to discourage that reading.

3 Results

3.1 Study population

The complete-case analysis cohort comprised 88,427 individuals aged 6–24 with a first depression diagnosis between 2006 and 2019 (after excluding 5,368 patients with missing parental education, family income, or family history; see Figure 1). Of these, 41,185 (46.6%) initiated SSRI treatment within 28 days of diagnosis, while 47,242 (53.4%) did not. The mean age was 18 years ($SD = 3.5$), and 61.9% were female.

During the 12-week follow-up period, there were 1,801 suicidal behavior events (2.0% event rate), with 1,013 events among SSRI initiators and 788 events among non-initiators.

3.2 Baseline characteristics

Table 2 presents baseline characteristics by treatment group.

Table 2: Baseline characteristics by treatment group on the complete-case analysis cohort ($N = 88,427$, see Figure 1), before IPW weighting. SMD = absolute standardized mean difference. Baseline characteristics of the 5,368 patients excluded by the complete-case restriction are reported in Table 15.

Characteristic	SSRI Initiators ($N = 41,185$)	Non-initiators ($N = 47,242$)	SMD
<i>Demographics</i>			
Female, n (%)	26,527 (64.4)	28,203 (59.7)	0.10
Age category (years)			0.11

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(Table 2 continued)

Characteristic	SSRI Initiators (N = 41,185)	Non-initiators (N = 47,242)	SMD
6–11	736 (1.8)	1,552 (3.3)	
12–17	19,229 (46.7)	20,690 (43.8)	
18–24	21,220 (51.5)	25,000 (52.9)	
<i>Socioeconomic factors</i>			
Source of depression diagnosis			0.12
Outpatient	37,840 (91.9)	41,772 (88.4)	
Inpatient	3,240 (7.9)	5,200 (11.0)	
Other/unknown source	105 (0.3)	270 (0.6)	
Parental education			0.08
Primary	1,900 (4.6)	2,942 (6.2)	
Secondary	18,739 (45.5)	21,939 (46.4)	
Post-secondary	20,546 (49.9)	22,361 (47.3)	
Family income			0.09
Negative	78 (0.2)	89 (0.2)	
Zero	46 (0.1)	91 (0.2)	
1st–20th percentile	7,072 (17.2)	9,553 (20.2)	
20th–80th percentile	25,099 (60.9)	28,551 (60.4)	
>80th percentile	8,890 (21.6)	8,958 (19.0)	
<i>Family history (count of biological parents)</i>			
Suicidal behavior			0.02
Neither parent	37,247 (90.4)	42,386 (89.7)	
One parent	3,745 (9.1)	4,602 (9.7)	
Both parents	193 (0.5)	254 (0.5)	
Depression			0.03
Neither parent	30,838 (74.9)	34,837 (73.7)	
One parent	9,395 (22.8)	11,181 (23.7)	
Both parents	952 (2.3)	1,224 (2.6)	
<i>Prior psychiatric diagnoses</i>			
Suicidal behavior, n (%)	2,876 (7.0)	4,663 (9.9)	0.10
Overdose/poisoning, n (%)	2,151 (5.2)	3,776 (8.0)	0.11
Anxiety disorder (F40, F41), n (%)	4,052 (9.8)	4,281 (9.1)	0.03
Stress/adjustment disorder, n (%)	3,708 (9.0)	5,362 (11.4)	0.08
ADHD, n (%)	4,893 (11.9)	7,239 (15.3)	0.10
Substance use disorder, n (%)	1,454 (3.5)	3,244 (6.9)	0.15
Alcohol use disorder, n (%)	1,998 (4.9)	3,222 (6.8)	0.08
OCD, n (%)	1,229 (3.0)	1,148 (2.4)	0.03

(continued on next page)

(Table 2 continued)

Characteristic	SSRI Initiators (N = 41,185)	Non-initiators (N = 47,242)	SMD
Sleep disorder, n (%)	1,502 (3.6)	1,880 (4.0)	0.02
Anorexia nervosa, n (%)	1,138 (2.8)	1,223 (2.6)	0.01
Bulimia nervosa, n (%)	457 (1.1)	381 (0.8)	0.03
Autism spectrum disorder, n (%)	2,308 (5.6)	2,463 (5.2)	0.02
Conduct disorder, n (%)	672 (1.6)	1,094 (2.3)	0.05
Bipolar disorder, n (%)	289 (0.7)	846 (1.8)	0.10
Psychotic disorder, n (%)	400 (1.0)	847 (1.8)	0.07
Organic mental disorder, n (%)	0 (0.0)	0 (0.0)	0.00
Cluster B personality disorder, n (%)	259 (0.6)	666 (1.4)	0.08
Intellectual disability, n (%)	542 (1.3)	635 (1.3)	0.00
<i>Prior medications</i>			
Antipsychotic, n (%)	1,299 (3.2)	1,489 (3.2)	0.00
Hypnotic/sedative, n (%)	11,299 (27.4)	6,939 (14.7)	0.32
Benzodiazepine, n (%)	1,823 (4.4)	856 (1.8)	0.15
Psychostimulant, n (%)	2,368 (5.7)	3,028 (6.4)	0.03
Opioid, n (%)	645 (1.6)	908 (1.9)	0.03
Antiepileptic, n (%)	121 (0.3)	194 (0.4)	0.02
Mood stabilizer, n (%)	351 (0.9)	980 (2.1)	0.10
Addiction medication, n (%)	118 (0.3)	99 (0.2)	0.02
<i>Healthcare utilization</i>			
Psychiatric hospitalization, n (%)	8,007 (19.4)	12,177 (25.8)	0.15
<i>SSRI type (initiators only)</i>			
Sertraline, n (%)	18,751 (45.5)		
Fluoxetine, n (%)	13,703 (33.3)		
Escitalopram, n (%)	4,733 (11.5)		
Citalopram, n (%)	3,702 (9.0)		
Paroxetine, n (%)	284 (0.7)		
Fluvoxamine, n (%)	12 (0.0)		

3.3 Intention-to-treat analysis

Figure 2 presents the weighted cumulative incidence curves for suicidal behavior by treatment group in the intention-to-treat analysis. These curves are derived from weighted Kaplan-Meier estimates as $1 - \hat{S}(t)$.

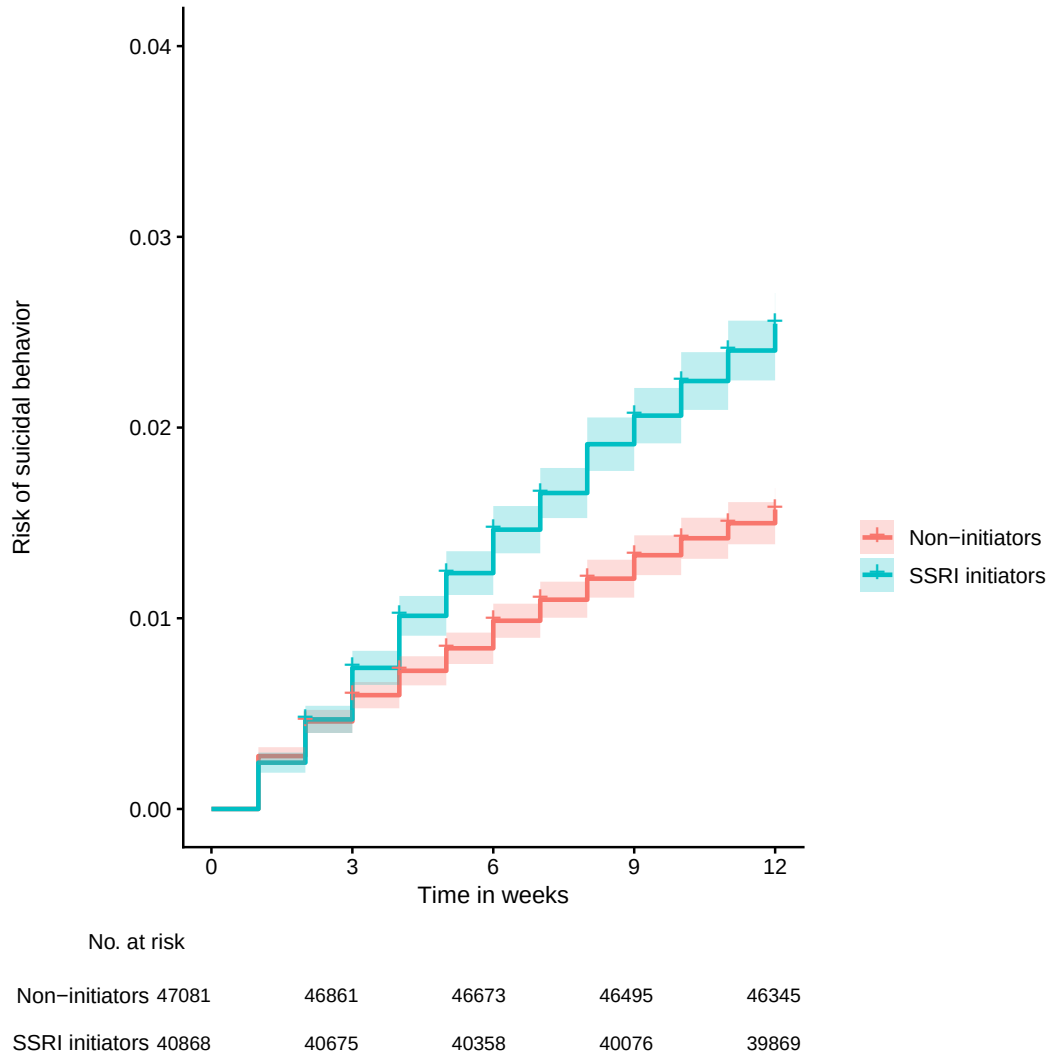


Figure 2: Weighted cumulative incidence curves for suicidal behavior by SSRI treatment status (intention-to-treat analysis), derived from Kaplan-Meier estimates. Shaded areas represent 95% confidence intervals. At-risk counts in the panel below the curves are IPW-weighted (matching the weighted Kaplan-Meier estimator).

Table 3 summarizes the 12-week intention-to-treat results, including the raw patient counts in each arm. The at-risk panel beneath Figure 2 reports IPTW-weighted counts and so falls a few hundred patients below these raw totals.

Table 3: Intention-to-treat analysis of SSRI initiation on 12-week suicidal behavior risk. Risk = Adjusted 12-week cumulative incidence. RD = Adjusted risk difference in percentage points (pp). RR = Adjusted risk ratio. “Adjusted” = IPTW-weighted.

	N	Events	Risk (%)	RD (pp, 95% CI)	RR (95% CI)
SSRI initiators	41,185	1,013	2.54	+0.97 (0.78, 1.17)	1.62 (1.47, 1.78)
Non-initiators	47,242	788	1.57		

3.4 Sensitivity analyses

3.4.1 Exclusion of inpatient diagnoses

The main analysis was repeated on the cohort with inpatient index diagnoses excluded ($n = 79,987$). Table 4 compares the restricted-cohort estimate with the main full-cohort estimate.

Table 4: Sensitivity analysis: ITT estimate with and without inpatient index diagnoses

Cohort	N	Events	RD (pp, 95% CI)	RR (95% CI)
Main (full cohort)	88,427	1,801	+0.97 (0.78, 1.17)	1.62 (1.47, 1.78)
Excluding inpatients	79,987	1,330	+0.92 (0.74, 1.11)	1.75 (1.57, 1.96)

Excluding inpatients yielded a slightly smaller absolute risk difference (RD shifted from +0.97 $\rightarrow$ +0.92 pp) but a slightly larger risk ratio (RR 1.62 $\rightarrow$ 1.75). Both estimates remain clearly significant and consistent in direction. The pattern is consistent with the lower baseline risk in the restricted cohort: removing the high-risk inpatient subset reduces the absolute scale of the contrast while the proportional effect of SSRI initiation is preserved. This indicates the headline finding is not driven by the inpatient subset.

3.4.2 Grace period

The 14-day rebuild reclassifies 4,077 patients (those whose first dispensation fell in the (14, 28]-day window) from initiators to non-initiators rather than excluding them. Of those, 3,870 remain in the complete-case analysis cohort. After complete-case filtering, the 14-day cohort size is essentially identical to the 28-day cohort (88,426 vs. 88,427). The single-patient discrepancy is a small-numbers artifact of re-running the matching procedure, not a structural effect of the shorter grace window. Under each rule, every non-initiator is assigned a baseline by random sampling from the distribution of time-from-diagnosis-to-prescription among initiators. Switching to the 14-day rule changes that distribution, so every non-initiator’s assigned baseline date changes. For one patient the new date happens to fall on or after their death or emigration date, so they no longer meet the requirement to be alive and resident in Sweden at baseline. Table 5 compares the rebuilt 14-day analysis with the main 28-day analysis.

Table 5: Sensitivity analysis: ITT estimate under 28-day (main) and 14-day grace periods

Grace period	N	Events	RD (pp, 95% CI)	RR (95% CI)
28 days (main)	88,427	1,801	+0.97 (0.78, 1.17)	1.62 (1.47, 1.78)
14 days	88,426	1,807	+0.79 (0.58, 0.99)	1.46 (1.33, 1.61)

The 14-day cutoff produced smaller estimates (RD $+0.97 \rightarrow +0.79$ pp, RR $1.62 \rightarrow 1.46$). Both estimates remain clearly significant and in the same direction. The reduction reflects the 3,870 reclassified patients in the analysis cohort (initiators under the 28-day rule whose first SSRI dispensation fell 15–28 days after diagnosis, and non-initiators under the 14-day rule). They contributed events to the initiator group under the 28-day rule but to the non-initiator group under the 14-day rule, mechanically diluting the contrast when the grace window is shortened. Total events rise modestly ($1,801 \rightarrow 1,807$) for a related reason: under the 14-day rule the matching procedure draws non-initiator baselines from a distribution capped at 14 days rather than 28, so the assigned baselines are on average closer to the diagnosis date. Each non-initiator’s 12-week follow-up window therefore shifts earlier and covers more of the elevated suicidal-behavior risk in the weeks immediately after diagnosis. The headline finding is robust in direction and remains clinically meaningful in magnitude across grace-period specifications.

3.4.3 Missing-indicator method

To assess how much the complete-case restriction (Section 2.6) drives the headline result, we re-fit the 12-week ITT analysis on the full eligible cohort ($N = 93,795$, including the 5,368 patients excluded under the complete-case rule). A single binary `any_miss` indicator ($=1$ for patients with sentinel-99 in at least one of the four partially-observed covariates) was added to the PS model, and every patient with `any_miss` = 1 was forced to the modal reference level on all four covariates so the indicator coefficient absorbs the entire missing-stratum contribution. Table 6 compares the missing-indicator estimate with the main complete-case estimate.

Table 6: Sensitivity analysis: ITT estimate under the complete-case (main) and missing-indicator cohorts

Cohort policy	N	Events	RD (pp, 95% CI)	RR (95% CI)
Complete case (main)	88,427	1,801	+0.97 (0.78, 1.17)	1.62 (1.47, 1.78)
Missing indicator	93,795	1,948	+0.98 (0.79, 1.18)	1.61 (1.47, 1.76)

Recovering the 5,368 patients dropped by the complete-case rule barely shifted the estimate (RD $+0.97 \rightarrow +0.98$ pp, RR $1.62 \rightarrow 1.61$). This indicates the headline finding is not driven by selection on observed missingness in the four partially-observed covariates. The method’s known biases (Section 4.4) mean this should be read as a robustness check rather than an unbiased estimate on the full cohort.

3.5 Heterogeneous treatment effect analysis

3.5.1 Heterogeneity test

The calibration test from the initial causal forest yielded a p-value of 0.021, providing strong evidence of treatment effect heterogeneity across baseline covariates. This justified proceeding with the iterative causal forest algorithm to identify patient subgroups with differential treatment effects.

3.5.2 Variable importance

The initial causal forest identified 12 variables with importance above the mean threshold, which were retained for subgroup identification. The top variables identified as potential effect modifiers were:

1. Prior suicidal behavior diagnosis (importance: 0.127)
2. Sex (importance: 0.110)
3. Prior overdose/poisoning (importance: 0.103)
4. Age group (importance: 0.097)
5. Family income (importance: 0.081)
6. Parental education level (importance: 0.074)
7. Healthcare source (importance: 0.069)
8. Alcohol use disorder (importance: 0.057)
9. Prior psychiatric hospitalization (importance: 0.049)
10. Family history of suicidal behavior (importance: 0.045)
11. Hypnotic medication (importance: 0.043)
12. Family history of depression (importance: 0.034)

Figure 3 presents the variable importance scores for the top 20 covariates.

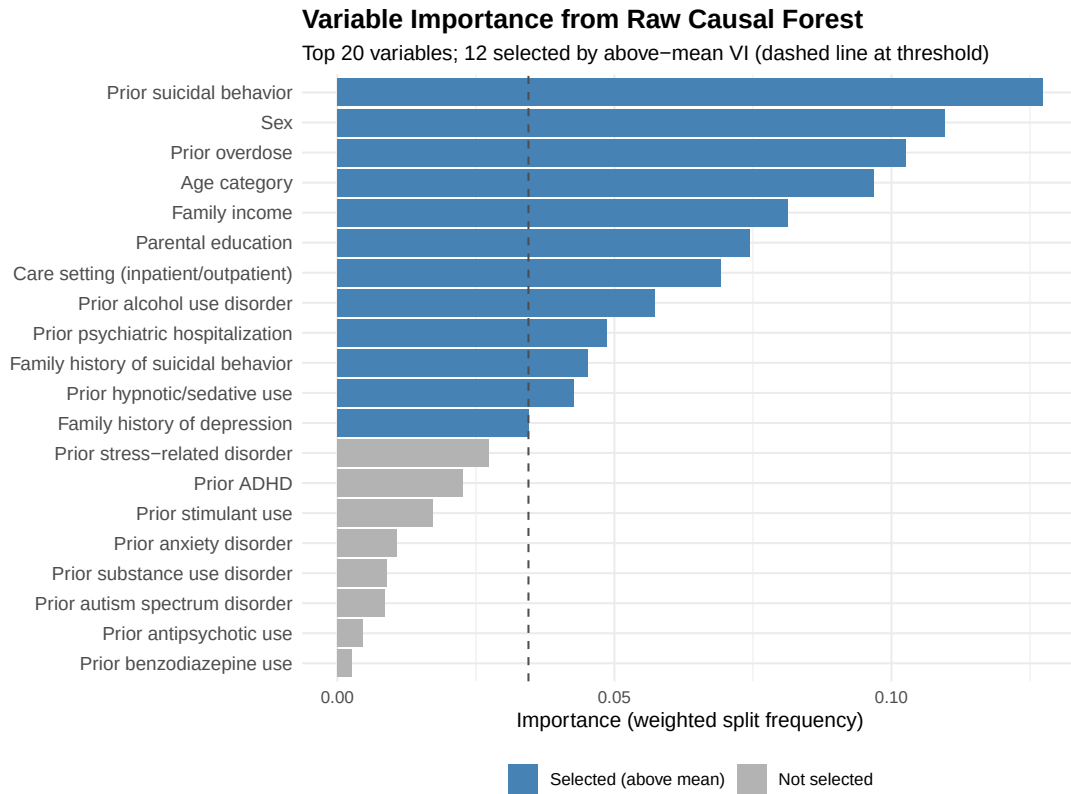


Figure 3: Variable importance from the initial causal forest, showing the top 20 variables ranked by their contribution to treatment effect heterogeneity. Importance is measured as the weighted frequency of splits on each variable.

3.5.3 Individual-level treatment effect heterogeneity

Figure 4 shows the distribution of individual-level CATE estimates from the initial (raw) causal forest, sorted from highest to lowest. The majority of patients have small positive CATEs near the average treatment effect. A minority on the left of the curve have substantially elevated predicted risk increases following SSRI initiation. The dashed line indicates the average treatment effect across the full cohort.

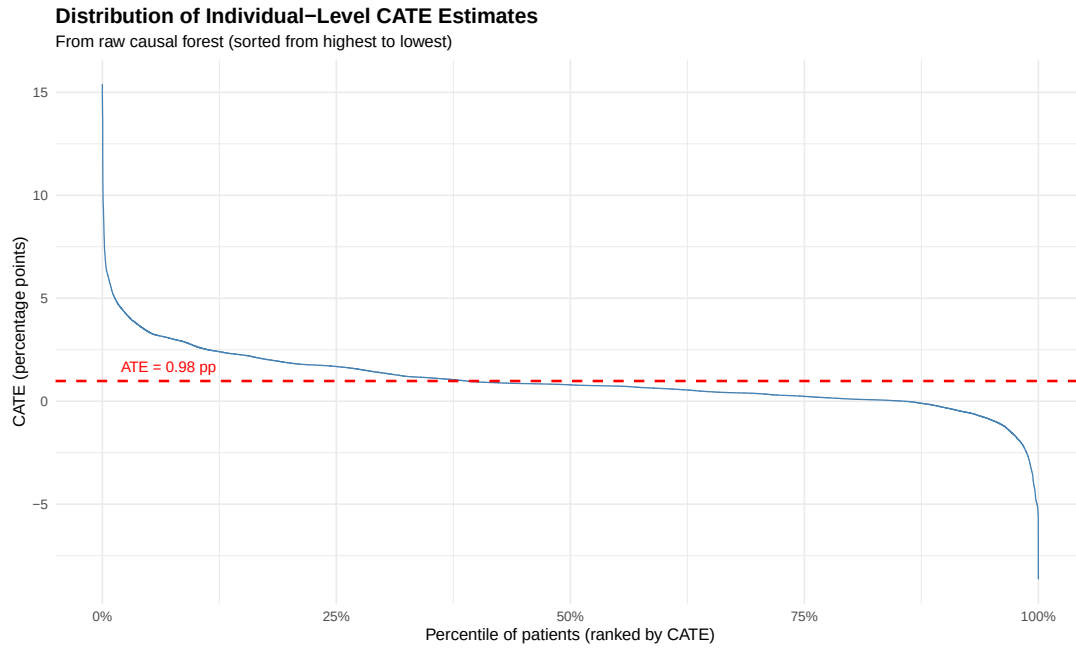


Figure 4: Distribution of individual-level conditional average treatment effect (CATE) estimates from the raw causal forest, sorted from highest to lowest. Each point represents one patient. The dashed line indicates the overall average treatment effect.

Figure 5 shows the same distribution as a histogram, with kernel density overlay, providing a complementary view of how the predicted CATEs are spread across the cohort.

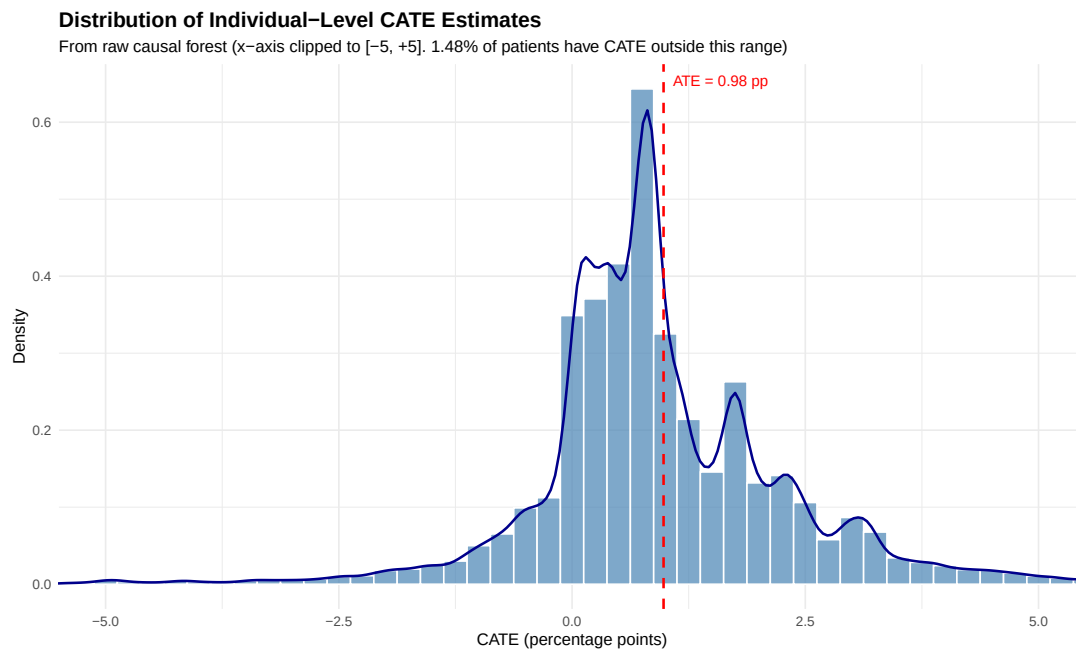


Figure 5: Histogram of individual-level CATE predictions from the raw causal forest, with kernel density overlay. The dashed line indicates the overall average treatment effect.

3.5.4 CATE calibration

To assess whether the causal forest’s individual-level CATE predictions are well-calibrated, we binned patients into deciles of predicted CATE and compared the mean predicted effect with the IPTW-weighted observed treatment effect within each bin (Figure 6). A well-calibrated model produces points along the diagonal (predicted = observed). For comparison, we also computed calibration for two alternative CATE estimators on the same cohort: a doubly-robust learner (DR) and a T-learner.

Calibration is summarized by three statistics: the slope of the regression of observed bin-means on predicted bin-means (ideal = 1), the intercept (ideal = 0), and the mean absolute calibration error (MACE) across the ten bins. Table 7 reports these statistics for the three estimators alongside the predicted and observed decile-mean ranges.

Table 7: Calibration statistics for the three CATE estimators (causal forest, doubly-robust learner, T-learner). Slope, intercept, and R^2 are from the regression of observed bin-means on predicted bin-means (ideal slope = 1, intercept = 0, $R^2 = 1$). MACE = mean absolute calibration error in percentage points (ideal = 0). Predicted and observed ranges are the decile-mean min/max across the ten bins.

Estimator	Slope	Intercept	R^2	MACE (pp)	Predicted (pp)	Observed (pp)
Causal forest	0.21	0.73	0.500	0.82	−1.32 to 3.86	0.40 to 1.51
Doubly-robust	0.09	0.85	0.215	1.59	−4.19 to 7.11	0.18 to 1.63
T-learner	2.89	−1.91	0.952	1.44	−1.14 to 4.07	−6.70 to 10.00

The three estimators fail calibration in different ways. The causal forest has the lowest MACE (0.82 pp), but its slope of 0.21 is well below 1. Observed decile-means span only 0.40 to 1.51 pp around the cohort ATE while predicted decile-means range from −1.32 to 3.86 pp, so the CF overstates how much treatment effect varies across the cohort. For context, a trivial estimator that always predicts the cohort ATE (+0.97 pp) would have MACE roughly equal to the mean distance from the observed deciles to the ATE – well below 0.82 pp – so the CF’s MACE is mostly explained by its mean tracking the ATE on average, not by capturing heterogeneity. The T-learner has by far the best rank-ordering ($R^2 = 0.952$) – useful for ranking patients by treatment effect – but its observed decile range (−6.70 to 10.00 pp) is implausibly wide for the cohort-size and event-rate combination, suggesting propensity-weight instability in tail deciles rather than informative heterogeneity. The doubly-robust learner is nearly flat in slope (0.09) and poor in rank-ordering ($R^2 = 0.215$), reflecting the high variance of its pseudo-outcomes in this cohort.

CATE Calibration: Predicted vs Observed Treatment Effects

Patients binned into deciles of predicted CATE; error bars = 95% bootstrap CI. Dashed grey: perfect calibration ($y = x$). Solid colored: linear fit with 95% CI band.

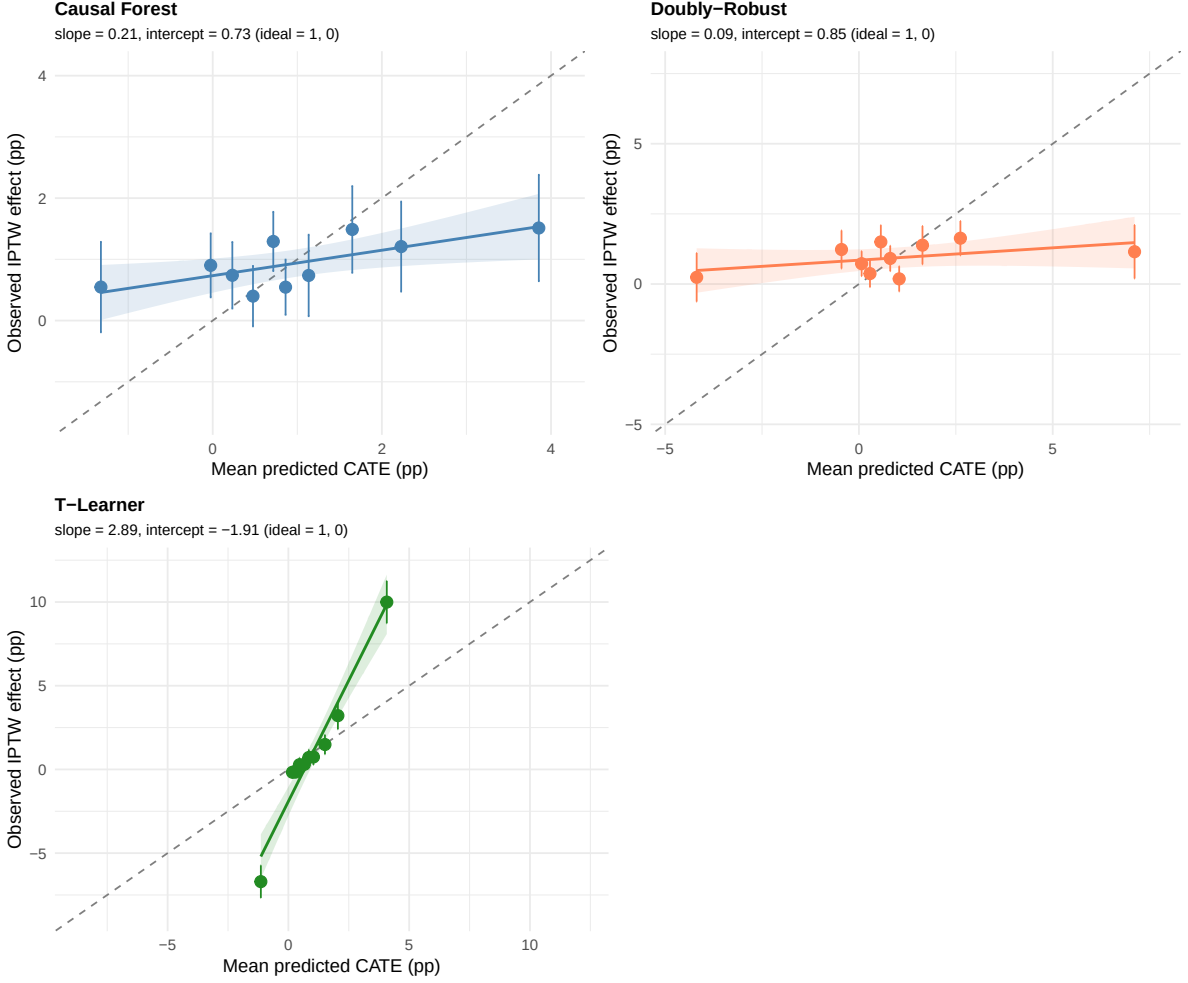


Figure 6: CATE calibration plot. Patients are binned into deciles of predicted CATE; within each bin, the mean predicted CATE (x-axis) is compared to the IPTW-weighted observed treatment effect (y-axis). Error bars represent 95% bootstrap confidence intervals. The dashed line indicates perfect calibration.

3.5.5 Subgroup identification

Given the heterogeneity-test rejection (Section 3.5.1), we ran the iCF algorithm to identify the tree structure underlying it. The Wang 2024 iCF selects depth by cross-validated mean squared error (CV-MSE), growing iCFs at each candidate depth $D \in \{2, 3, 4, 5\}$ and reporting the argmin. Table 8 reports the per-depth CV-MSE, within-fold standard deviation, and the variables appearing in the voted tree at each depth.

Table 8: Cross-validation MSE and within-fold standard deviation by candidate depth. The CV-argmin (depth 4) and the 1-SE-selected headline depth (depth 2) are both indicated. “Splitting variables” lists the variables appearing in the voted tree at that depth, in depth-of-appearance order (root first).

Depth	Mean CV MSE	Within-fold SD	Splitting variables
2	0.092833	0.004128	Sex
3	0.092837	0.004123	Sex, age
4	0.092786	0.004118	Sex, age, education
5	0.092819	0.004124	Age, sex, fh-depr, income, education

The CV-argmin is depth 4, but the CV-MSE landscape is flat: the within-fold SD exceeds the across-depth range by roughly two orders of magnitude, so the argmin is governed by simulation variance rather than out-of-sample fit, and all four candidate depths sit inside the 1-SE band around the minimum. By the 1-SE rule of Breiman et al. [27], we report the shallowest depth in the plateau, with deeper trees in Section 3.5.8 as sensitivity. The rule selects depth 2 (two subgroups).

Figure 7 visualizes the headline depth-2 tree while Table 9 reports its per-subgroup characteristics and treatment effects.

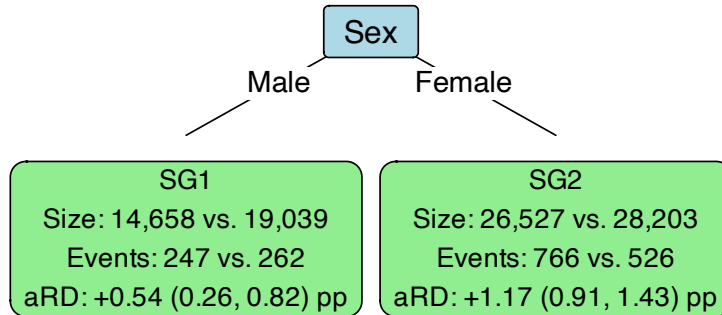


Figure 7: Headline iCF decision tree (depth 2, selected by the 1-SE rule). Decision nodes (blue) show the splitting variable and threshold; leaf nodes (green) show the identified subgroups with their CATE estimates in percentage points. Deeper voted trees are shown in Section 3.5.8.

Table 9: Subgroup-specific treatment effects identified by iterative causal forest. CATE = conditional average treatment effect estimated using IPTW within each subgroup. Rates are events per 100 person-periods at 12 weeks.

Subgroup	N	SSRI Rate (%)	Control Rate (%)	CATE (pp, 95% CI)
Male	33,697	1.69	1.38	+0.54 (0.26, 0.82)
Female	54,730	2.89	1.87	+1.17 (0.91, 1.43)

The iCF CATE is +0.54 pp in males and +1.17 pp in females (roughly a $2.2\times$ difference) and both subgroup confidence intervals exclude zero.

3.5.6 Stratified analysis: sex differences

To validate the iCF subgroup findings, we conducted a stratified IPW analysis within each identified subgroup. Patients were assigned to iCF subgroups based on the synthetic decision tree, and the standard IPW-weighted Kaplan-Meier analysis (identical to the main ITT analysis) was run separately within each subgroup. Table 10 compares the iCF CATE estimates with the independent stratified IPW risk differences.

Table 10: Validation: iCF CATE vs. stratified IPW risk difference (RD). iCF CATE uses IPTW with causal forest propensity scores. Stratified IPW RD uses IPW with logistic regression propensity scores (same method as the main ITT analysis). All values in percentage points.

Subgroup	iCF CATE (95% CI)	Stratified IPW RD (95% CI)
Male	+0.54 (0.26, 0.82)	+0.53 (0.24, 0.81)
Female	+1.17 (0.91, 1.43)	+1.27 (1.00, 1.53)

The two independent estimation approaches produce closely aligned point estimates and confidence intervals. The iCF CATE and the stratified IPW RD agree to within 0.1 pp in each subgroup, and both place the female effect at roughly $2.2\times$ the male effect. This suggests that the headline subgroup contrast is not an artifact of the causal forest methodology.

3.5.7 Stratified analysis: prior suicidal behavior

Prior suicidal-behavior diagnosis ranked first in the raw causal forest’s variable importance (Figure 3: importance 0.127, ahead of sex at 0.110). Yet, it did not appear in the voted tree at any candidate depth (Table 8). To check whether this discrepancy hides a real effect modifier that the plurality vote failed to converge on, we ran the standard IPW-weighted ITT analysis. The analysis is identical to the main analysis, except that the prior-suicidality covariate is dropped from the PS formula within each stratum where it is constant. The analysis was done separately in the 80,888 patients without and the 7,539 patients with a prior suicidal-behavior diagnosis at baseline. Table 11 reports each stratum’s sample size, event count, and 12-week cumulative incidence by treatment arm, and Table 12 the resulting risk differences and risk ratios.

Table 11: Per-stratum sample size, event count, and 12-week cumulative incidence of suicidal behavior by treatment arm (IPW-weighted Kaplan–Meier).

Stratum	N	Events	Non-initiators (%)	Initiators (%)
No prior suicidal behavior	80,888	1,253	1.12	2.03
Prior suicidal behavior	7,539	548	6.36	8.27

Table 12: ITT effect estimates stratified by prior suicidal-behavior diagnosis at baseline.

Stratum	RD (pp, 95% CI)	RR (95% CI)
No prior suicidal behavior	+0.91 (0.73, 1.09)	1.82 (1.62, 2.04)
Prior suicidal behavior	+1.91 (0.60, 3.21)	1.30 (1.09, 1.54)

The two strata show a familiar absolute-versus-relative split: patients with a prior suicidal-behavior diagnosis have a higher baseline risk (Table 11) and a numerically larger absolute risk increase under SSRI initiation (RD +1.91 vs. +0.91 pp), but a smaller relative risk (RR 1.30 vs. 1.82). The 95% CIs of the two RDs overlap, so the absolute-scale difference is not itself statistically distinguishable. This is discussed further in Section 4.2.2.

3.5.8 Per-depth decision trees

To examine what additional subgroup structure emerges at deeper tree depths inside the 1-SE plateau, we grew iCF forests at each candidate depth and extracted the voted decision tree with CATE estimates. Figures 8–14 show the voted trees at target depths 2 through 5. “Target depth” is the `maxdepth` parameter passed to `grf`. It is possible for the plurality vote across iCF iterations to settle on a structure shallower than the target, in which case the rendered tree is shallower than its caption’s target value. Depth 2 (Figure 8) is the headline tree reported in Section 3.5.5. Depth 4 is the CV-argmin (Figure 12). The cohort is complete-case for all categorical covariates, so all ordinal splits are between legitimate covariate categories.

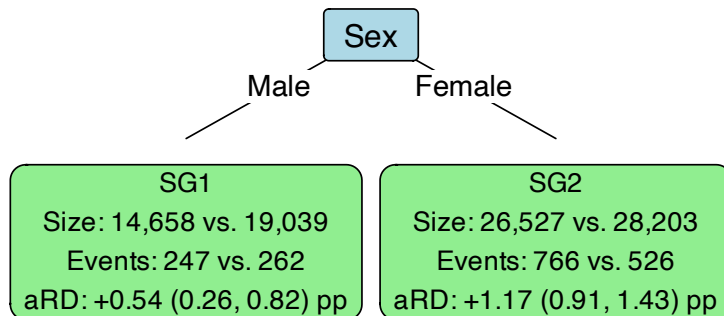


Figure 8: Voted iCF tree at target depth 2 (2 subgroups, voted-tree stability 1.00).

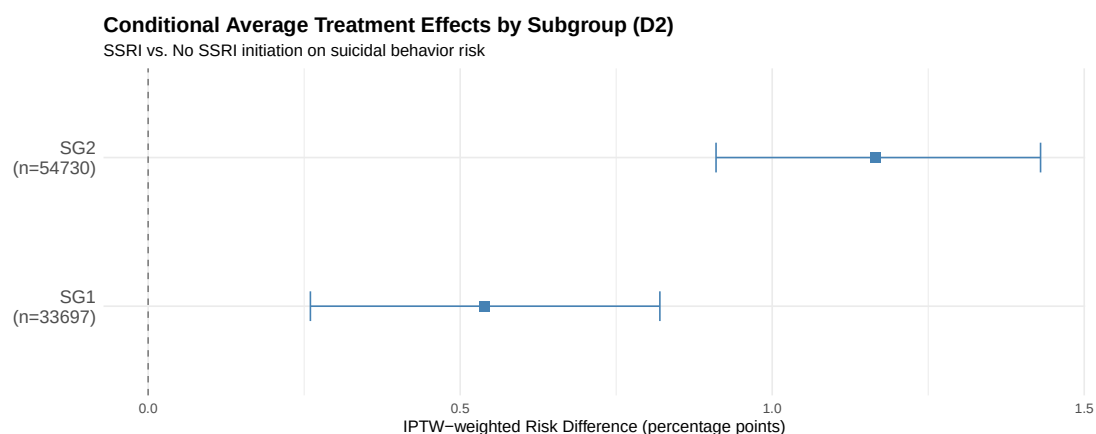


Figure 9: Subgroup CATEs at target depth 2 (IPTW-weighted risk differences).

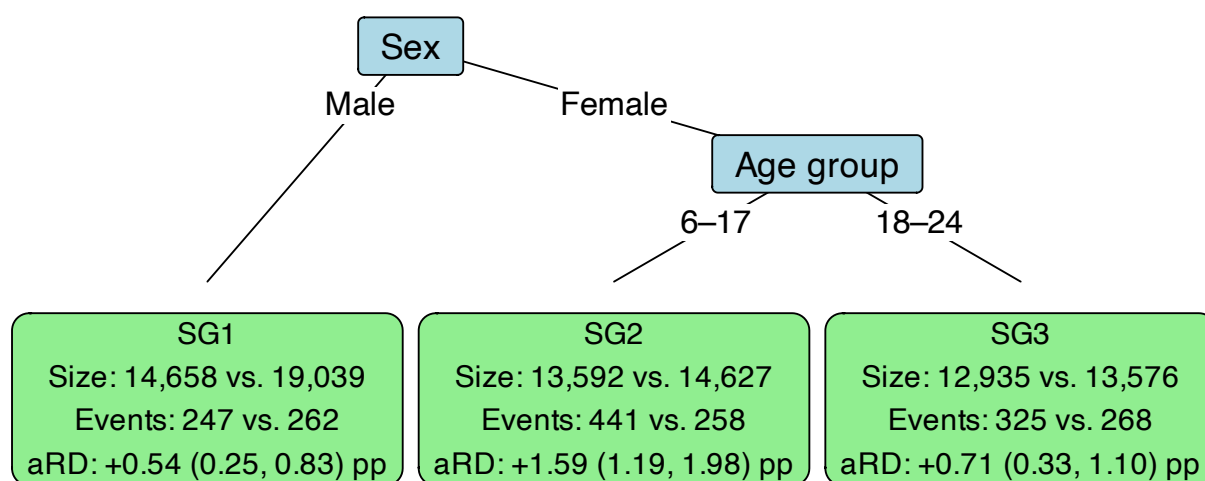


Figure 10: Voted iCF tree at target depth 3 (3 subgroups, voted-tree stability 0.46).

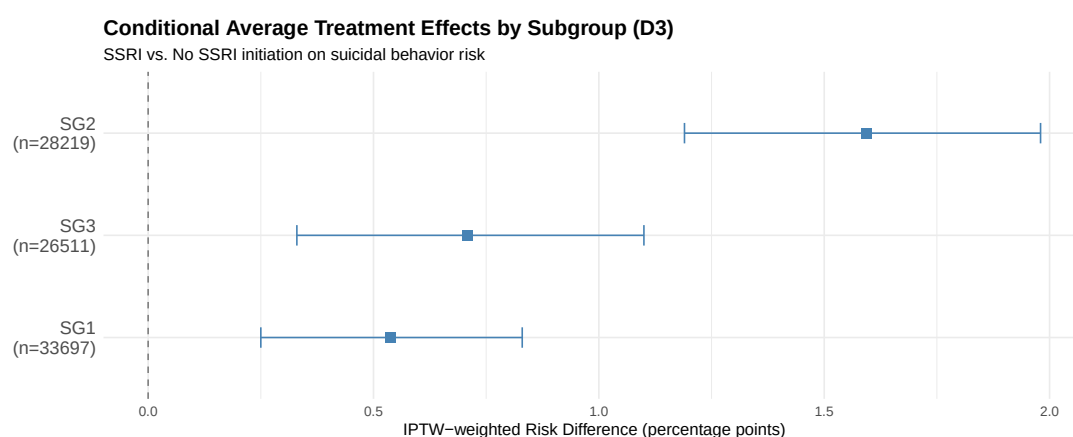


Figure 11: Subgroup CATEs at target depth 3 (IPTW-weighted risk differences).

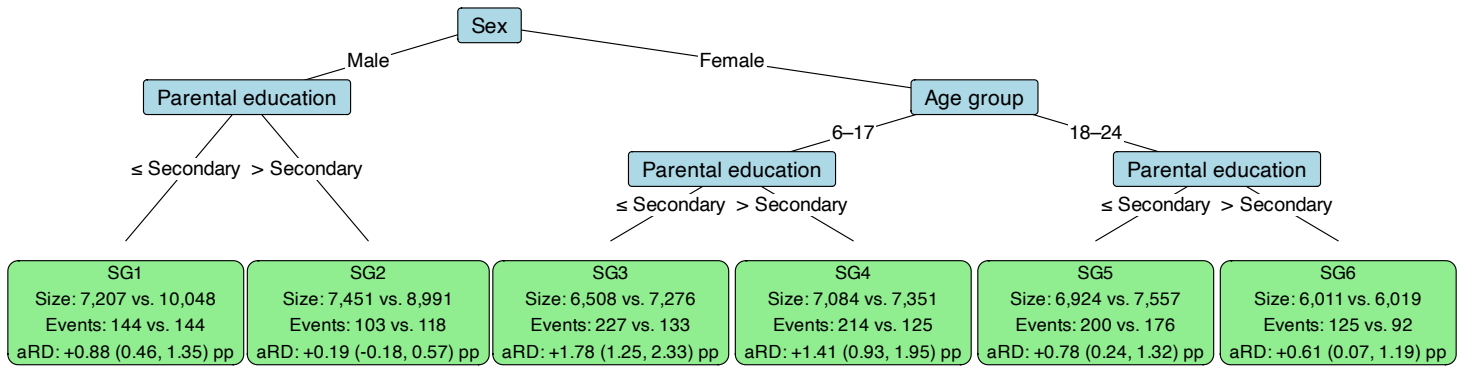


Figure 12: Voted iCF tree at target depth 4 (6 subgroups, voted-tree stability 0.27).

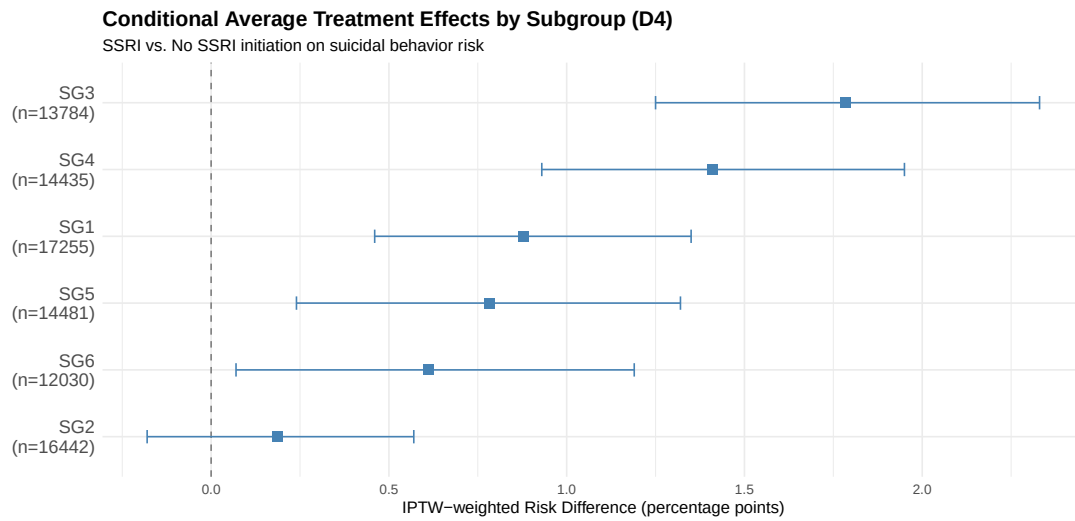


Figure 13: Subgroup CATEs at target depth 4 (IPTW-weighted risk differences).

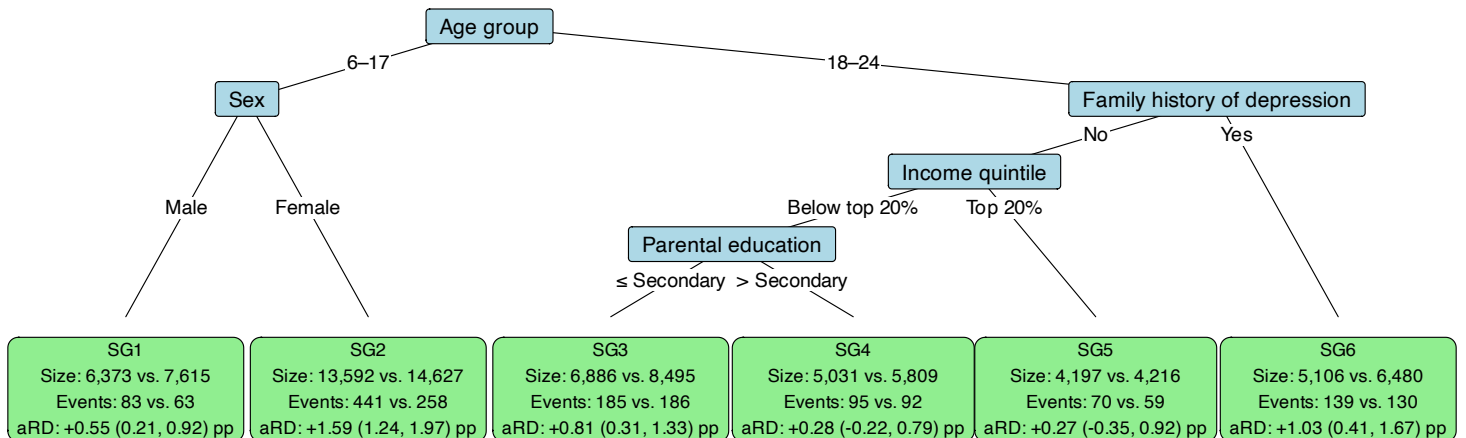


Figure 14: Voted iCF tree at target depth 5 (6 subgroups, voted-tree stability 0.64).

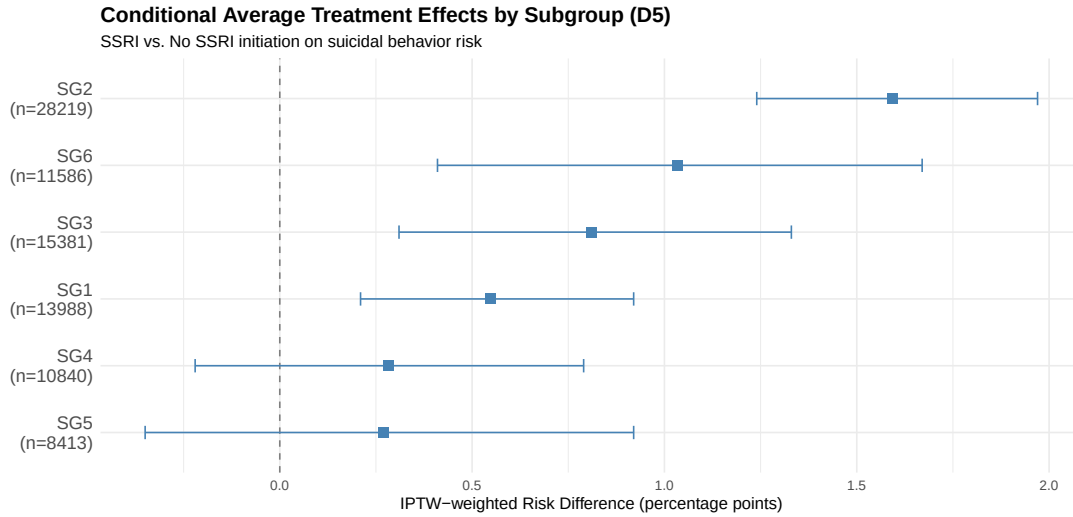


Figure 15: Subgroup CATEs at target depth 5 (IPTW-weighted risk differences).

Figure 14 shows the D5 leaf-level aRDs directly. Two of the six subgroups have CIs crossing zero, consistent with the D5 tree being underdetermined at this cohort size.

3.5.9 Voted-tree structure distribution

The voted-tree stability summary in Section 3.5.8 reports a single plurality fraction per depth. Figure 16 unpacks that fraction by showing the top-5 most-frequent tree structures across iCF iterations at each candidate depth. At depth 2 the algorithm picks a single structure (D1:Sex) unanimously across all iterations. Depth 3 narrows to 3 close candidates with the top vote at 0.46. Depth 4 fragments across 14 distinct structures with the top vote falling to 0.27. Depth 5 partially recovers (3 structures, top vote 0.64) on an age-rooted tree that no longer matches the D2 sex split.

Voted-tree structure distribution by candidate depth

Top 5 most frequent tree structures per depth. Voted (most popular) structure in blue.

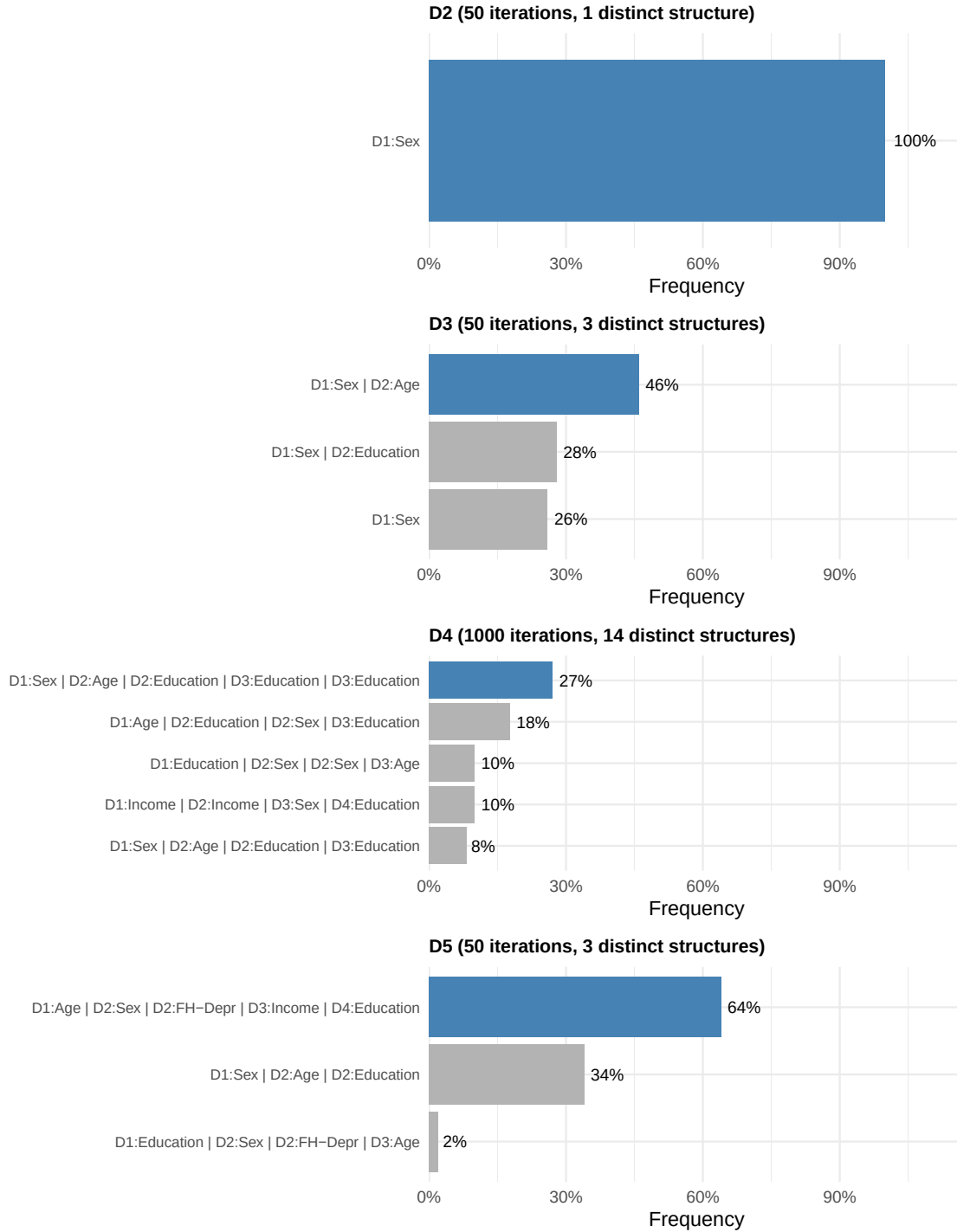


Figure 16: Voted-tree structure distribution across iCF iterations at each candidate depth. Each row in a panel is a distinct tree structure encoded as a pipe-separated list of depth-indexed split variables. Bar length is the fraction of iCF iterations that converged on that structure. The voted (top-plurality) structure is shown in blue. D4 uses 1,000 iterations (the headline-detection budget). D2/D3/D5 use 50 iterations each (the sensitivity budget).

The shape of Figure 16 is the structural counterpart of the flat CV-MSE plateau in Table 8: when the candidate depth is shallow enough to be identified by the data, the iCF iterations converge on a single tree. When it is deeper than the data supports, they scatter across many roughly-equivalent partitions.

Notably, the frequency of the most commonly selected tree falls sharply with depth but recovers partially at D5. D5’s recovery is on a different structure where the first split is on age and sex drops to depth 2. This is a depth-dependent re-ordering rather than a contradiction, and consistent with Wang et al. [24]’s observation that deeper voted trees can settle on different structures than their shallow counterparts.

3.5.10 High-dimensional iCF (hdiCF)

To probe for effect modifiers beyond the curated covariate set, we ran the hdiCF pipeline with 39 automatically generated HD features (9 inpatient diagnoses, 21 outpatient diagnoses, 9 prescriptions) alongside the curated variables. Per Wang 2025, the calibration-based heterogeneity test was bypassed (the sparse ordinal HD features dilute its signal), and the pipeline proceeded directly to variable selection and subgroup identification. The CV-MSE landscape across candidate depths was flat (relative range $\sim 10^{-4}$, far below the within-fold SD of $\sim 10^{-2}$), so all four depths fell inside the 1-SE band. Applying the same 1-SE rule used for base iCF (Section 3.5.5), the headline depth is depth 2 (2 subgroups: male and female), with voted-tree stability 1.00. The Wang-2024 CV-argmin would have selected depth 4. That deeper tree is presented as a sensitivity in Section 3.5.11. Of the 7 variables retained by the top-10% importance filter, one was an HD feature: inpatient depressive episode (`dx.inp.F32`).

The 7 retained variables, in descending importance, are:

1. Prior suicidal behavior (importance: 0.106)
2. Prior overdose/poisoning (importance: 0.086)
3. Family income (importance: 0.073)
4. Parental education (importance: 0.065)
5. Inpatient: F32 (importance: 0.062)
6. Sex (importance: 0.058)
7. Age group (importance: 0.054)

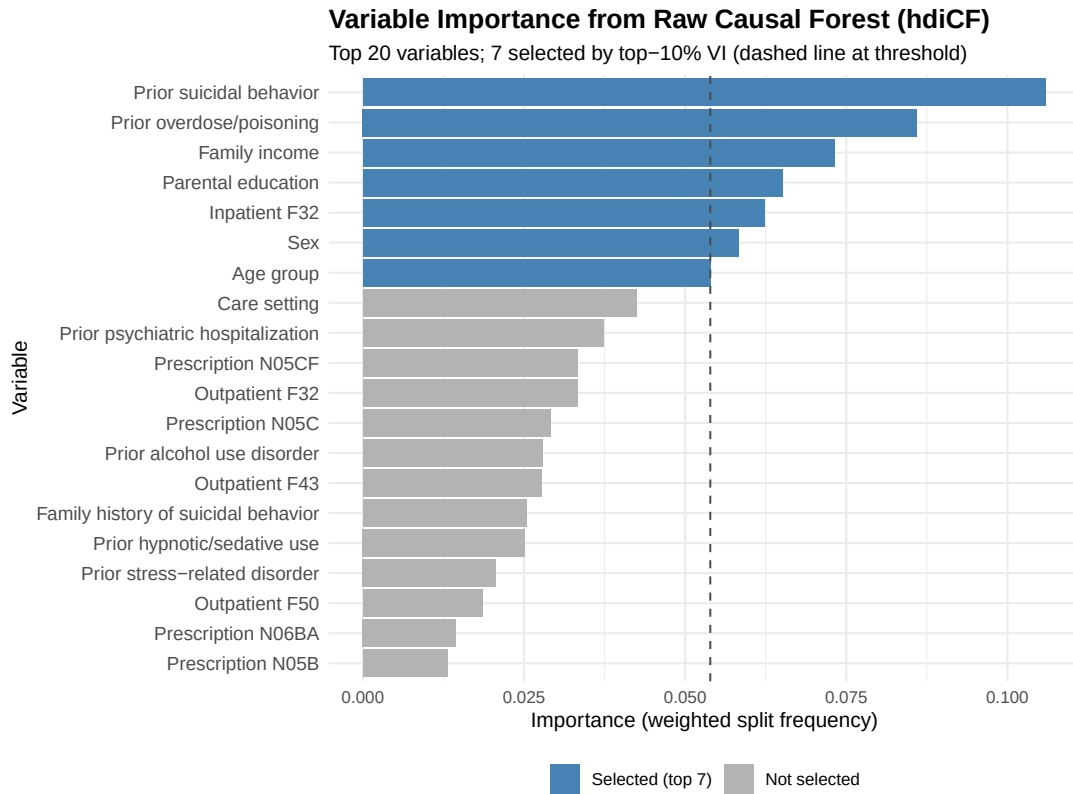


Figure 17: Variable importance from the raw causal forest in the hdiCF pipeline.

The headline tree (Figure 18) is the depth-2 sex split, with stability 1.00 across iCF iterations. Stratified IPW within each leaf reproduces the iCF CATEs to within 0.1 pp, mirroring the base-iCF validation result.

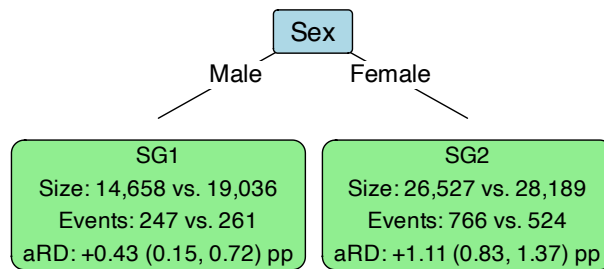


Figure 18: Headline hdiCF decision tree (depth 2, selected by the 1-SE rule).

3.5.11 Per-depth decision trees (hdiCF)

As with the base iCF (Section 3.5.8), we extracted the voted decision tree at each candidate target depth to examine what additional structure emerges as the hdiCF tree is allowed to grow deeper. Figures 19–25 show the four voted trees. Target depth 2 (Figure 19) is the headline while target depth 4 (Figure 23) is the CV-argmin.

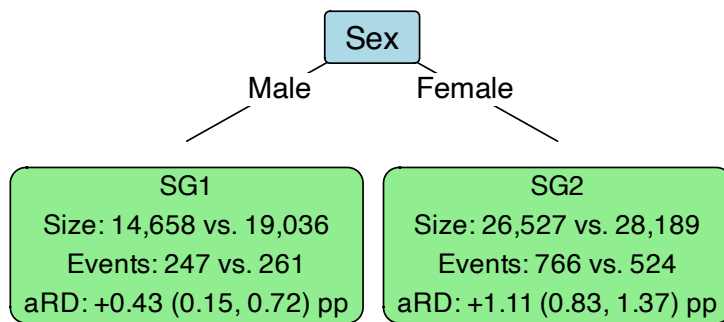


Figure 19: hdiCF voted tree at target depth 2 (2 subgroups).

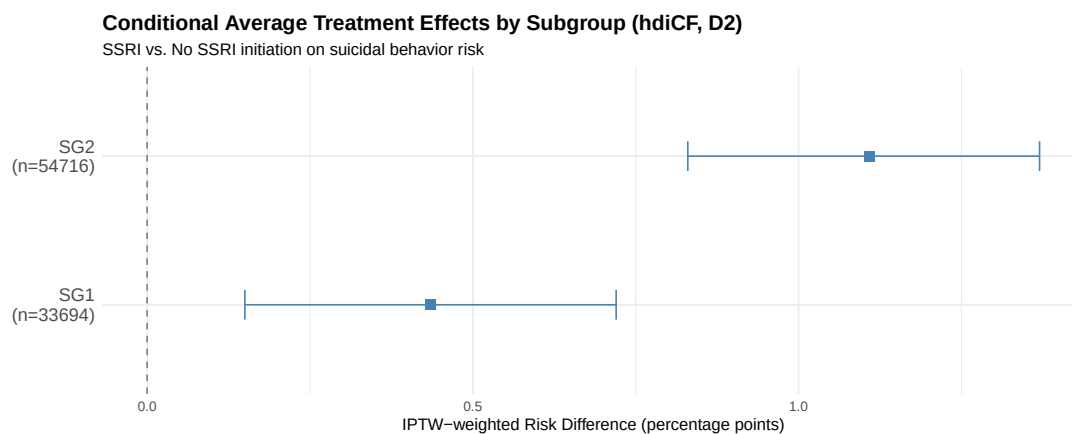


Figure 20: hdiCF subgroup CATEs at target depth 2 (IPTW-weighted risk differences).

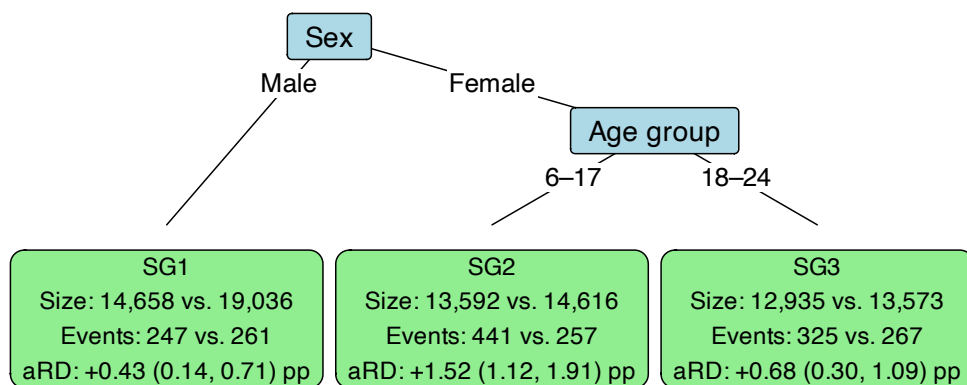


Figure 21: hdiCF voted tree at target depth 3 (3 subgroups).

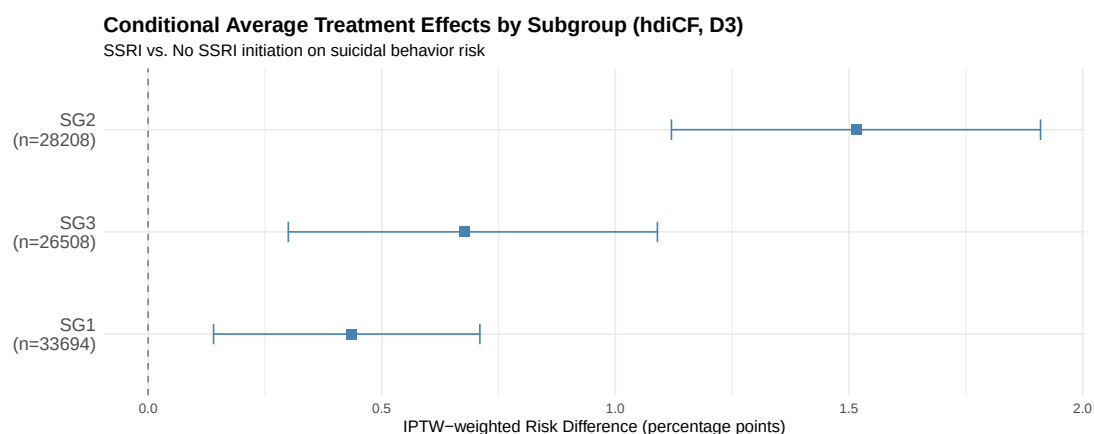


Figure 22: hdiCF subgroup CATEs at target depth 3 (IPTW-weighted risk differences).

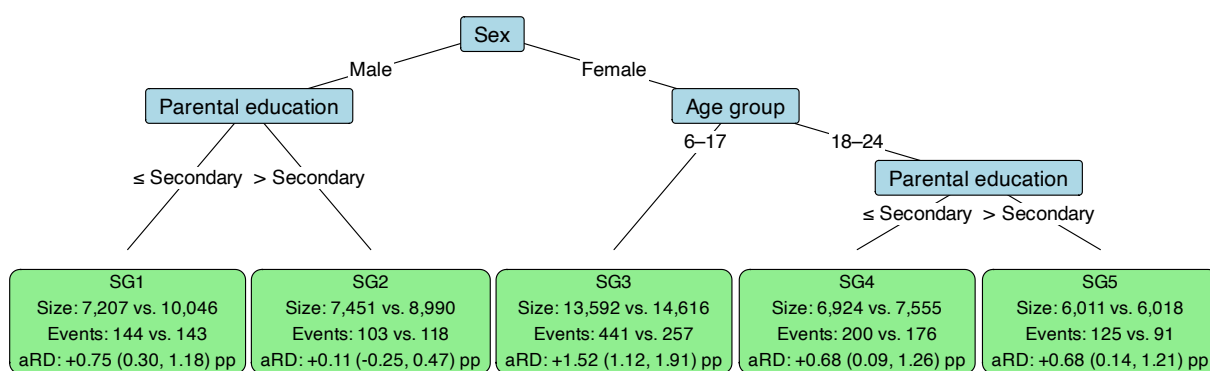


Figure 23: hdiCF voted tree at target depth 4 (5 subgroups).

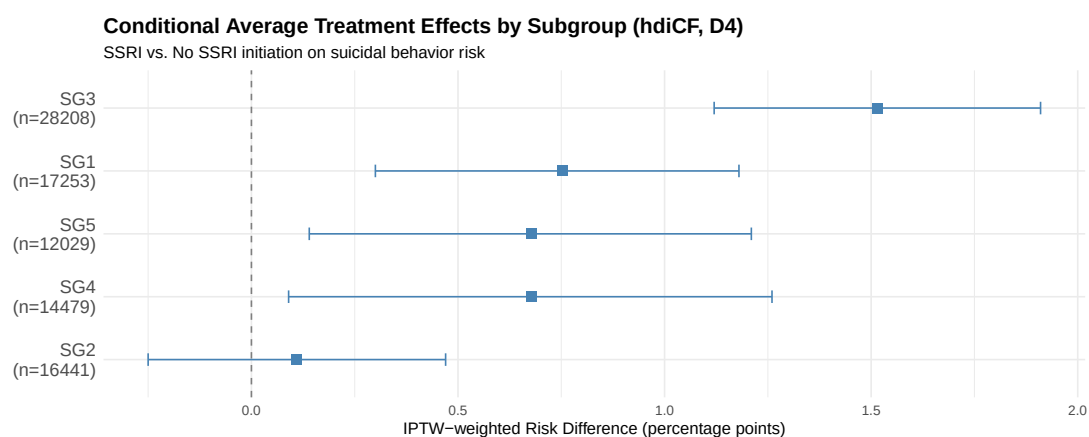


Figure 24: hdiCF subgroup CATEs at target depth 4 (IPTW-weighted risk differences).

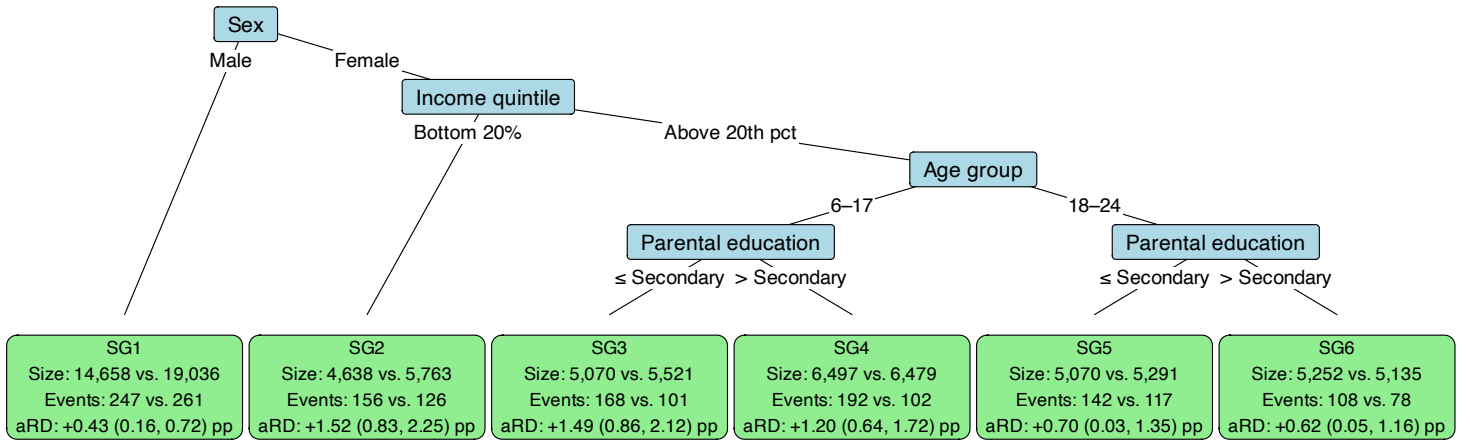


Figure 25: hdiCF voted tree at target depth 5 (6 subgroups).

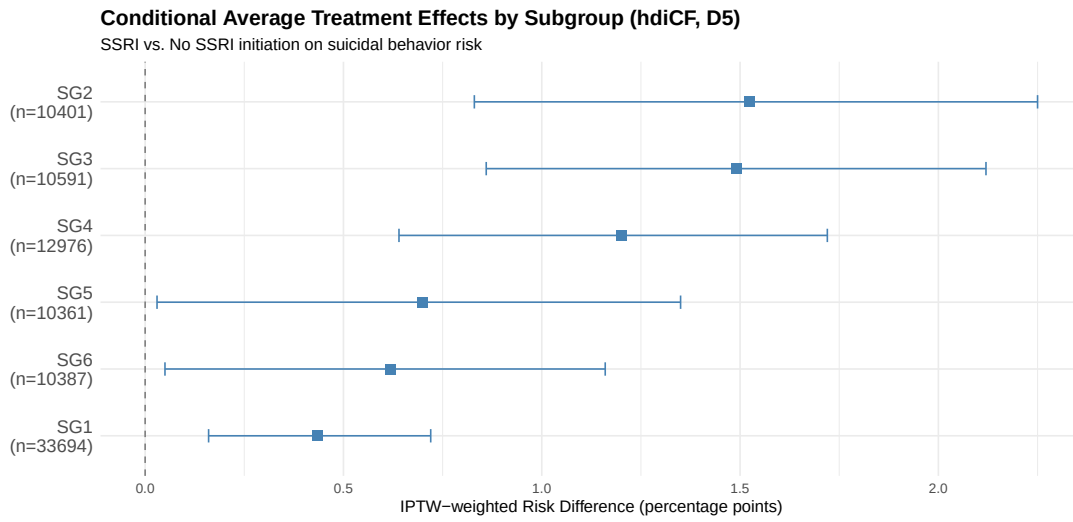


Figure 26: hdiCF subgroup CATEs at target depth 5 (IPTW-weighted risk differences).

3.5.12 Voted-tree structure distribution (hdiCF)

Figure 27 unpacks the per-depth voted-tree stability summary in the same way as Figure 16 did for base iCF. At depth 2 the hdiCF iterations converge unanimously on D1:Sex (1 distinct structure). Depth 3 narrows to 3 close candidates with the top vote at 0.70. Depth 4 fragments across 12 distinct structures with the top vote at 0.20. Depth 5 retains only 2 candidate structures and chooses between them with top vote 0.54. Notably, the HD feature (Inpatient F32) does not appear in any of the structures.

Voted-tree structure distribution by candidate depth

Top 5 most frequent tree structures per depth. Voted (most popular) structure in blue.

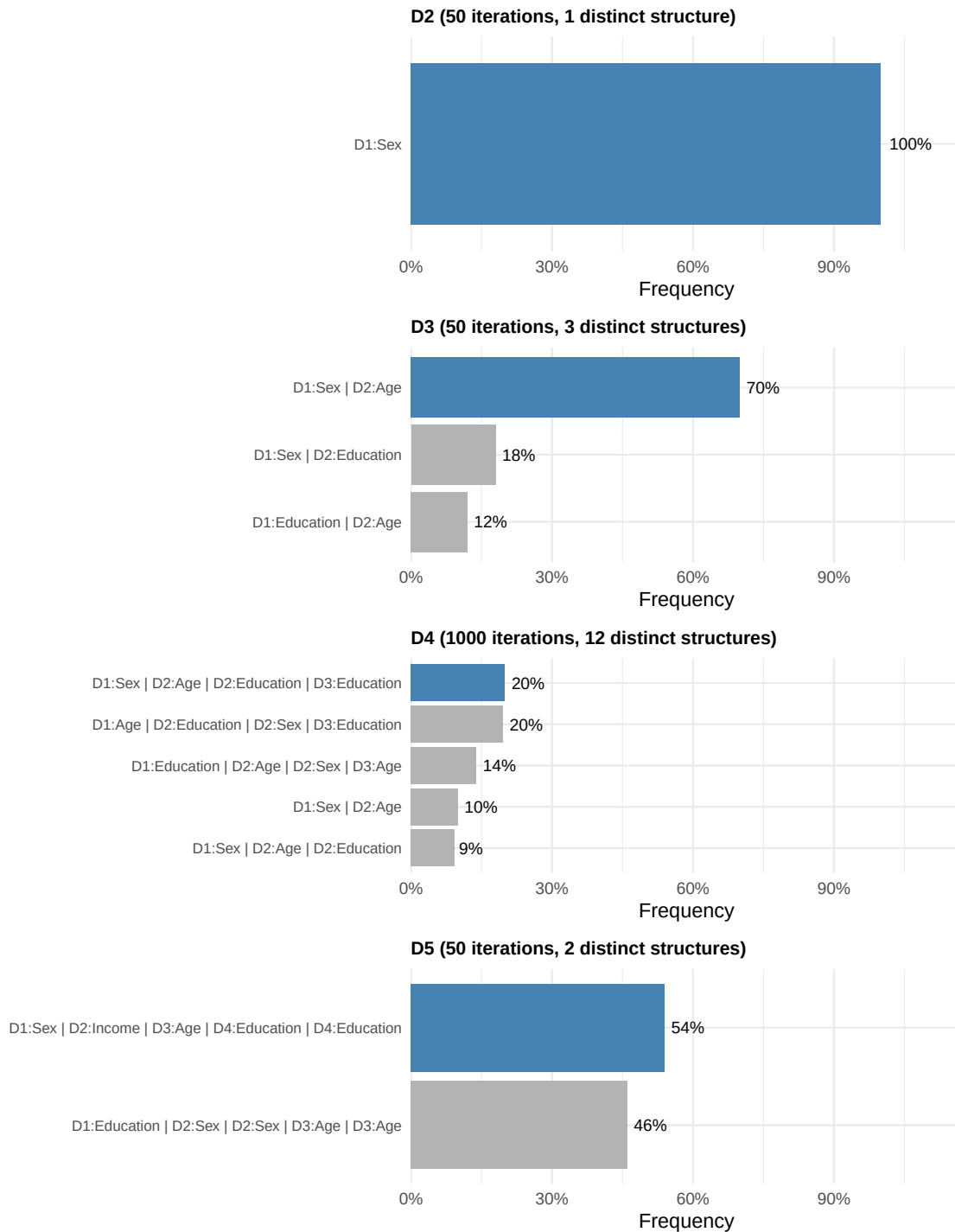


Figure 27: Voted-tree structure distribution across hdiCF iterations at each candidate depth. Bar length is the fraction of iCF iterations that converged on that structure. The voted (top-plurality) structure is shown in blue. D4 uses 1,000 iterations. D2/D3/D5 use 50 iterations each.

3.5.13 Missing-indicator iCF/hdiCF triage

To rule out that missingness in the four CCA covariates carries detectable HTE signal the headline iCF and hdiCF are blind to, we reran step 1 (raw causal forest + variable importance) of each pipeline on the full eligible cohort with a single binary `any_miss` indicator added as a candidate splitting variable (Section 3.4.3). The indicator’s variable importance was 0.0070 in iCF (mean VI 0.0286) and 0.0041 in hdiCF (mean VI 0.0137). This is well below the above-mean selection threshold in both pipelines, so `any_miss` could not survive variable selection and could not appear in any voted tree at any depth.

3.6 Targeting operator characteristic and Qini curves

To evaluate the clinical value of the causal forest’s individual-level CATE predictions for treatment targeting, we computed Qini curves following [22]. The Qini curve plots the expected number of suicidal behavior events that would be prevented by withholding SSRI from the patients predicted to be most harmed (ranked by CATE), compared to a random withholding policy. The area under the curve (AUTOC) summarizes the overall gain as a single number. $AUTOC = 0$ means CATE-based ranking is no better than random.

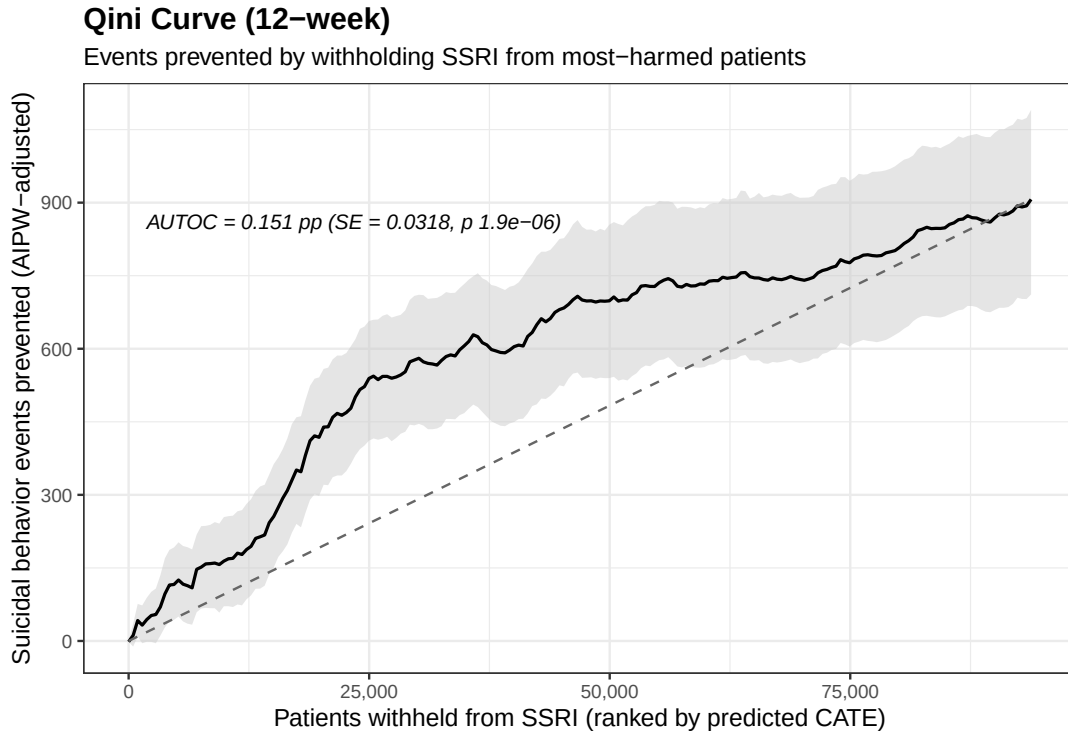


Figure 28: Qini curve at 12-week follow-up using individual-level CATE predictions from the base iCF causal forest. Solid line: CATE-based targeting policy. Dashed line: random-policy baseline. Shaded region: 95% bootstrap CI.

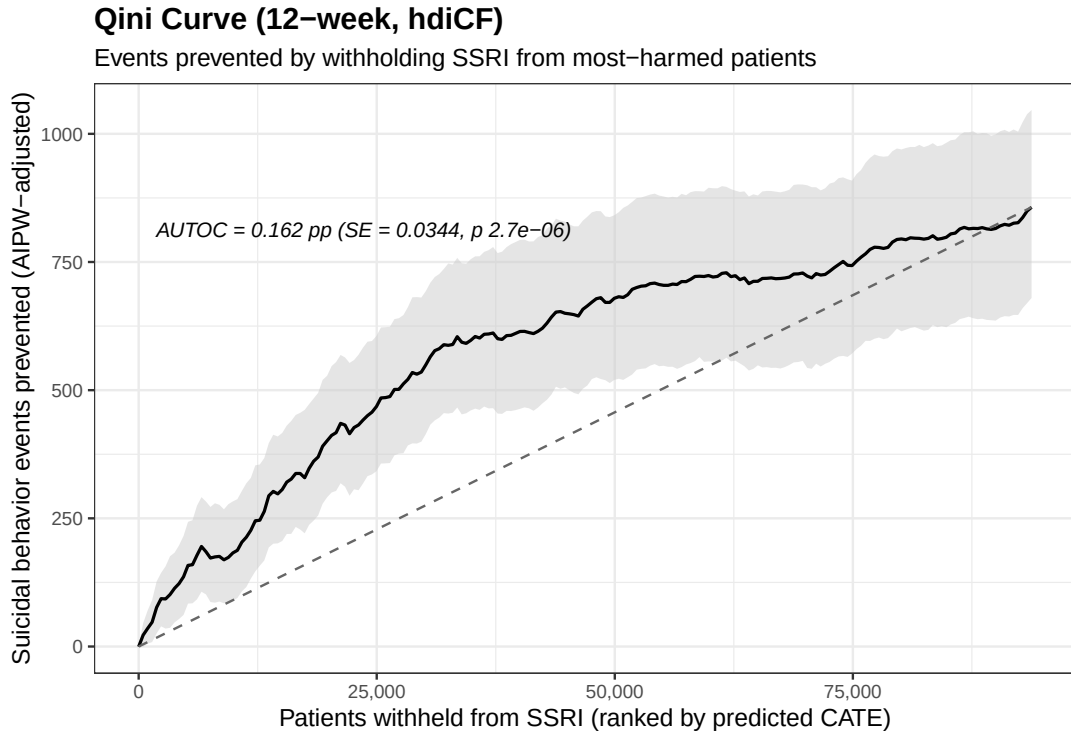


Figure 29: Qini curve at 12-week follow-up using individual-level CATE predictions from the hdiCF causal forest.

The base-iCF AUTO C was 0.151 pp (95% bootstrap CI 0.092, 0.216; $p = 1.91 \times 10^{-6}$). The hdiCF AUTO C was 0.162 pp (0.095, 0.229; $p = 2.70 \times 10^{-6}$). Both AUTO Cs are positive, so per-patient ranking captures some real heterogeneity. Both are also small relative to the overall ATE (+0.97 pp), because most of the heterogeneity is the binary sex contrast that the headline tree already shows. Hence, ranking patients by predicted CATE adds little on top of just stratifying by sex.

4 Discussion

4.1 Main analysis

In a national cohort of 88,427 Swedish individuals aged 6–24 with a first depression diagnosis, ITT SSRI initiation within 28 days was estimated to increase the 12-week absolute risk of suicidal behavior by +0.97 pp (95% CI 0.78, 1.17), corresponding to a relative risk of 1.62 (1.47, 1.78), under the identification assumptions of the target trial emulation (no unmeasured confounding given the baseline covariates in Table 1, positivity, and consistency). The effect was larger in ages 6–17 (RD +1.26 pp) than in 18–24 (RD +0.70 pp). The headline estimate was robust to excluding inpatient index diagnoses (+0.92 pp) and to a 14-day grace period (+0.79 pp).

The closest comparator is Lagerberg et al. [9], a target-trial emulation of a Stockholm-

County cohort (162,267 individuals aged 6–59, depression diagnoses 2006–2018). They reported an overall ITT RD of 0.15 pp (RR 1.50), with age-specific RDs of 1.48 pp (RR 2.90) for ages 6–17 and 0.27 pp (RR 1.59) for ages 18–24, and no effect at age 25+. Our age-specific estimates match in direction and are broadly comparable in magnitude. Our larger overall RD (+0.97 pp vs. 0.15 pp) reflects the cohort restriction to ages 6–24, whereas Lagerberg’s overall estimate is diluted by the much larger and effectively null 25–59 segment. Methodological differences (national vs. Stockholm-only cohort, data through 2019 vs. 2018, Kaplan–Meier with stabilized IPW vs. pooled logistic regression) are unlikely to drive the broad agreement.

The broader RCT evidence is consistent in direction. A meta-analysis of pediatric antidepressant trials reported roughly a two-fold increase in suicidal ideation or attempts [8], the trigger for the FDA’s 2004 black-box warning summarized in Hammad et al. [6]. Our cohort-based 12-week RD falls in the same direction. The main contribution here is extending the trial-based signal to a population-level emulation under current real-world prescribing.

4.2 Subgroup analyses

4.2.1 Sex differences

The iCF heterogeneity test confirmed that the effect is not uniform across patients ($p = 0.021$). We found that sex is the main covariate that changes the SSRI effect. The 1-SE-selected headline tree was a single sex split: CATE +0.54 pp in males (0.26, 0.82) and +1.17 pp in females (0.91, 1.43), a roughly $2.2\times$ difference. We confirmed this in two ways. Firstly, re-estimating each leaf’s effect with a separate method (stratified IPW) reproduced the iCF CATEs to within 0.1 pp. Secondly, the per-patient targeting analysis (AUTOC = 0.151 pp, $p = 1.91 \times 10^{-6}$) was positive but small relative to the overall +0.97 pp ATE. Hence, the binary sex contrast already captures most of the heterogeneity, and ranking patients more finely by their predicted CATE adds little.

The high-dimensional augmentation in hdiCF preserved this sex split unchanged, with no HD feature stabilizing into any voted tree at any depth. The implications of this null methodological result for the second research question are discussed in Section 4.2.4.

On the sex axis itself, the most direct comparator is Lagerberg et al. [9]’s own sex-stratified analysis (Table S5), which we replicate methodologically. Their 12-week per-protocol results show a substantially elevated risk in females (overall RR 1.91, 95% CI 1.23, 2.94; RD 0.23 pp, 95% CI 0.05, 0.41), concentrated in the 6–17 stratum (RR 4.09, 95% CI 1.71, 9.80; RD 2.47 pp, 95% CI 0.19, 4.76, on 84 outcome events). Among males, the overall estimate is statistically inconclusive (RR 1.38, 95% CI 0.83, 2.27; RD 0.12 pp, 95% CI -0.09 , 0.33). The corresponding male 6–17 cell is uninformative (only 5 events

in initiators; RR CI 0.06, 49.82). Estimates are near-null in both sexes from age 25 onwards. The directional pattern is consistent with our iCF result: the effect is larger in females in the pediatric stratum. The male pediatric CI cautions against reading the male point estimate as definitively null. Two caveats apply when making a comparison with Lagerberg’s results. First, Lagerberg’s sex-stratified analysis is per-protocol whereas our iCF analysis is based on ITT. Second, they do not report a formal sex–age interaction test.

Beyond the parent paper, evidence on sex-specific effects is sparse. The FDA’s 2004 black-box warning [10] and the pivotal pediatric meta-analyses [6, 8] stratified on age, not on sex, so sex-specific effect modification has not been a headline finding in that literature. The closest direct comparator is Termorshuizen et al. [28], a Dutch retrospective cohort study (66,196 patients starting antidepressants, of whom 5,636 were aged <25). This study reported an incidence rate ratio of 1.94 ($p = 0.065$) for suicide attempts in the first month of treatment among females <25, against 0.52 ($p = 0.368$) in males <25, and concluded that female sex was the modifier of early-attempt risk in young patients. Our analysis is consistent in direction, our cohort is larger, and we have employed a target-trial emulation rather than a before-versus-after comparison window. We therefore place a much sharper estimate around an effect that Termorshuizen’s smaller young-patient subgroup ($n = 5,636$) was statistically inconclusive about at the conventional 0.05 threshold.

Evidence from RCTs appears to be sparse. Khan et al. [29] pooled 15 placebo-controlled antidepressant trials (323 depressed outpatients, mean age ~ 42) and found that the SSRI effect size was roughly twice as large in women (0.82) as in men (0.44). SSRI responder rates were 64.8% in women versus 47.5% in men ($p < 0.05$). It should be noted that the cohort and endpoint are not the same as ours: Khan studied adults, and the endpoint was symptom response rather than suicidal behavior. However, the broader pattern emerging from the RCT-pooled and observational evidence is that sex matters on both sides of the risk–benefit calculation at SSRI initiation: the female response appears larger in both the therapeutic and the early-period harm direction. Our analysis reinforces the harm-side half of that pattern in a population-scale registry cohort of young people.

4.2.2 Prior suicidal behavior

Prior suicidal behavior was the strongest predictor of future suicidal behavior in this cohort by variable importance (Figure 3: importance 0.127, ahead of sex at 0.110). Despite this, it never appeared as a split in any voted iCF tree at any candidate depth. To check whether the iCF voting was missing a real effect modifier, we ran the IPW-weighted ITT estimate separately within the two prior-suicidality strata (Section 3.5.7). The two strata show an absolute-versus-relative split: patients with a prior suicidal-behavior diagnosis have a higher baseline risk, a numerically larger absolute risk increase under SSRI ini-

tiation, and a smaller relative risk, with overlapping 95% CIs on the RD scale. Both the stratified estimates and the iCF voting suggest that prior suicidal behavior is the strongest prognostic marker in this cohort but does not detectably modify the SSRI effect on the scale iCF uses to split (cross-validated MSE of the CATE). Variable importance in `grf` counts how often each variable gets used to split the data across the forest, with splits closer to the root weighted more heavily because they shape larger parts of the tree. The algorithm is designed so that these splits are driven by differences in treatment effect between subgroups, not by who has a higher baseline outcome risk overall. It achieves this by adjusting both the outcome and the treatment indicator for the covariates before any split is chosen. In practice this adjustment is never perfect, and some baseline-risk signal slips through. As a consequence, a variable can receive a high importance score either because it genuinely changes how patients respond to SSRI, or because it strongly predicts who is likely to have the outcome regardless of treatment. So when a top-ranked variable never appears as a split in any voted iCF tree, the natural interpretation is that it falls into the second category: a strong predictor of the outcome, but not an effect modifier.

Lagerberg et al. [9] reported the same null in this high-risk subgroup (per-protocol RR 1.17, 95% CI 0.58, 2.34 in patients with a history of suicidal behavior at baseline; their Table 3). They noted that the RCT evidence base for this group is thin because patients with a history of suicidal behavior are routinely excluded from antidepressant trials. The agreement between our stratified estimate, our iCF reading, and Lagerberg’s own stratified result reinforces a single conclusion: in a population-scale registry cohort, prior suicidal behavior identifies the patients at highest baseline risk but does not appear to alter the direction or magnitude of the SSRI-attributable risk increase on the relative scale.

4.2.3 Other subgroupings

The purpose of running iCF and hdiCF was to surface subgroup structure beyond what pre-specified demographics already capture, so the dominance of sex deserves a closer look. The variable-importance ranking retained 12 curated variables (Figure 3), a mix of demographics (sex, age group), socioeconomic variables (parental education, family income), prior-history flags (prior suicidal behavior, prior overdose, prior psychiatric hospitalization, prior alcohol use), and family-history indicators. Of these, only sex made it into the headline tree, and only age appeared as a deeper split the iCF iterations agree on. The clearest example of the variable-importance-vs-effect-modifier distinction is prior suicidal behavior, discussed in its own subsection above (Section 4.2.2). The same pattern holds for prior overdose and prior psychiatric hospitalization: both rank near the top of the variable-importance ranking but never enter a voted iCF tree at any candidate depth, suggesting they are prognostic markers of baseline outcome risk rather than modifiers

of the SSRI effect. “Most important by variable importance” and “most important effect modifier” are different questions with different answers: Figure 3 answers the first whereas Figure 7 answers the second.

Across the four candidate tree depths (Section 3.5.8), the per-depth voted trees elaborate the headline with age- and SES-based refinements. The depth-3 tree adds an age split within the female branch (CATE +1.59 pp at ages 6–17 vs. +0.71 pp at ages 18–24, Table 8), matching Lagerberg’s age finding [9]. Deeper trees (D4–D5) add candidate splits on parental education, family income, and family history. However, Figure 16 shows that the iCF iterations split their votes across 14 different D4 tree structures and 3 different D5 structures, with the most-popular D5 variant promoting age to the root and dropping sex to depth 2. We therefore read the deeper trees as exploratory rather than confirmed effect modifiers. To test whether any of the depth-3+ structures holds up, we would need to apply this method in a larger cohort where the cross-validated error is not flat across depth.

4.2.4 Methodological assessment of (hd)iCF in this cohort

The second research question asked whether high-dimensional iterative causal forests can uncover clinically meaningful subgroups in Swedish registry data. The pipeline did surface the binary sex contrast that replicates across an independent estimator and aligns with prior evidence (Section 4.2.1), so the data-driven HTE framework is not vacuous here. Two findings, however, temper the case for the high-dimensional augmentation specifically.

First, none of the 39 ordinally-encoded ICD-10 and ATC features generated by hdiCF stabilized into the voted tree at any candidate depth. The single HD feature that survived the top-10% importance filter (inpatient F32) never split a voted tree, and the depth-2 headline structure was identical to that of the base iCF (Section 3.5.10). In this cohort, hdiCF therefore added no information beyond what the curated covariate set already provided.

Second, the cross-validated MSE landscape was flat across all four candidate depths (relative range $\sim 10^{-4}$, well below the within-fold SD of $\sim 10^{-2}$), so the data did not uniquely identify a depth. We fell back on the CART 1-SE rule to select the shallowest stable structure, but the iterative procedure’s principal selling point — i.e. letting cross-validation choose tree depth — did not discriminate here. The structure-distribution analysis (Figure 16) shows the same flatness in tree space. The distribution is unanimous on depth 2 and dispersed across many roughly-equivalent partitions at depth 4.

The possible explanation is statistical power. With 1,801 events across 88,427 patients, the across-depth CV-MSE differences sit well below within-fold noise. A larger registry cohort, or pooled data across Nordic registries, would be the natural test of whether hdiCF

surfaces additional structure beyond sex. With the available data, the present analysis cannot resolve that question.

4.3 Strengths

- Register infrastructure: Complete national coverage of inpatient and outpatient diagnoses (NPR), all dispensed prescriptions (PDR), socioeconomic variables (LISA), and pseudonymized parental linkages (Multi-Generation Register). This gives us two strengths an ordinary clinical trial would not have. First, outcomes are captured passively in the registers as long as the patient is alive and resident in Sweden, so the cohort has essentially no loss to follow-up. Emigration and death are the only ways out and both are flagged in the registers. Second, family-history covariates are derived from each parent’s own NPR record rather than from participant self-report, yielding register-based, objective family-history measurements that trials almost never have.
- Target-trial framework: Explicit alignment of eligibility, treatment assignment, and time zero resolves immortal-time bias and structures the analysis as a transparent emulation of a hypothetical randomized trial [14, 15].
- Data-driven HTE analysis: The iCF and hdiCF pipelines replace the multiplicity of pre-specified subgroup tests with a single cross-validated heterogeneity test plus stable plurality-voted subgroups, controlling Type I error while still letting the data surface effect modifiers [22, 23].
- Validation via independent estimators: Two pre-registered sensitivity analyses (inpatient exclusion, 14-day grace period) and two independent CATE estimators (IPTW with causal-forest propensity scores, stratified logistic-regression IPW) provide validation of the headline estimate from different statistical methods.

4.4 Limitations

- Residual confounding: The IPW analysis assumes the propensity-score model captures all confounding by indication. Depression severity is only crudely proxied by the curated covariates (care setting, prior hospitalization, prior anxiety/stress diagnoses). We cannot rule out unmeasured confounders such as current suicidal ideation, which the NPR captures only when acted upon. The direction of any residual bias is ambiguous.
- Cohort restriction and generalizability: The 12-week follow-up window captures the period during which the suicidal-behavior signal is strongest in randomized trials but does not characterize longer-term effects. The cohort is restricted to patients with a

first depression diagnosis and no antidepressant in the prior year, so the contrast is between starting an SSRI and not starting one. Results do not generalize to SSRI use in patients with chronic or recurrent treatment histories.

- **Specialist-care-only cohort:** The Swedish National Patient Register covers inpatient and specialist-outpatient care but not primary care. Depression diagnoses managed entirely in primary care are therefore invisible to our cohort. Lagerberg et al. [9] additionally drew on Region Stockholm’s VAL database, which links primary-care diagnoses and so captures the milder cases that never reach specialist care. Our cohort is correspondingly biased toward more severe presentations, which may make the estimated effects less applicable to the larger segment of patients diagnosed and treated entirely in primary care.
- **Complete-case cohort:** Four covariates (parental education, family income, family history of depression, family history of suicidal behavior) had non-trivial missingness driven by incomplete registry coverage rather than by patient or clinician behavior. We restricted the analysis cohort to complete cases on these four variables, dropping 5,368 patients (5.72% of the post-washout cohort). Table 15 shows that the excluded patients differ from the retained cohort on several baseline characteristics, so the analysis is clearly not missing completely at random (MCAR). This incurs a selection bias.
- **Missing-indicator bias:** The missing-indicator sensitivity analysis (Section 3.4.3) recovers the dropped patients and yields essentially the same estimate, but should be read as a robustness check rather than an unbiased estimate on the full eligible cohort. When the underlying covariate is a confounder, the missing-indicator method is known to yield biased treatment-effect estimates even under MCAR [30], because the contrast within the “missing” stratum does not correspond to an exchangeable comparison.

4.5 Clinical implications

The reportable clinical signal is a binary sex difference in early-period suicidal-behavior risk at SSRI initiation: +0.54 pp (0.26, 0.82) in males and +1.17 pp (0.91, 1.43) in females, with the same contrast appearing at every candidate iCF depth. This adds a sex axis to the FDA black-box warning’s age-only stratification [10] and can reasonably be raised in a risk discussion at SSRI initiation. The +0.97 pp overall absolute increase corresponds to roughly one additional event per hundred patients treated over 12 weeks. This is small in absolute terms but clinically meaningful given the severity of the outcome. This risk difference must be balanced against the unmeasured benefits of SSRI treatment (symptom remission, prevention of chronic disease). The contrast is between starting an SSRI and

not starting one, not between SSRIs and an active alternative such as psychotherapy.

No subgroup we identified at any depth showed a protective effect of SSRI initiation on the suicidal-behavior endpoint at 12 weeks. The question is one of relative magnitude of harm, not of direction. Deeper subgroup distinctions (age in the female branch at depth 3, income and parental education at depths 4–5) are hypotheses to be tested against independent registry data and should not be taken as bases for current prescribing decisions.

4.6 Future research

- Severity-adjusted re-analysis: A planned follow-up publication will re-run the pipeline on a more recent version of the registry data, possibly based on the cohort in the Swedish Quality Register for Child and Adolescent Psychiatry (QBUP). This would allow the use of the SQD and CGAS severity measures for the propensity-score model and the iCF/hdiCF feature set.
- Replication of deeper iCF structure: Larger cohorts in which the CV-MSE landscape is not flat would let the iCF uniquely identify a depth and confirm or refute the candidate modifiers (age within females, SES gradients) surfaced at depths 3–5.
- Independent replication: Other Nordic registries, and non-Nordic cohorts with different prescribing patterns, could be used to test whether the sex contrast generalizes beyond Sweden.
- Longer follow-up: Lagerberg’s 52-week curves [9] suggest continued accumulation of events beyond the 12-week initiation window.
- Active comparator: An SSRI vs. psychotherapy (or watchful-waiting) emulation would address the clinical decision more directly than the start-vs-not-start contrast emulated here.

5 Conclusions

In a national cohort of Swedish youth with a first depression diagnosis, SSRI initiation within 28 days was estimated to increase the 12-week risk of suicidal behavior by +0.97 percentage points (95% CI 0.78, 1.17). The estimate was robust to inpatient-exclusion and 14-day grace-period sensitivity analyses and broadly aligned with prior target-trial-emulation evidence in the same age range. The iCF cross-validated MSE was flat across all candidate depths on this cohort. After applying the CART 1-SE rule, the data-driven analyses identified 2 subgroups defined by sex, with the female effect roughly 2.2× the male effect. Deeper iCF trees, presented as sensitivity analyses, suggested further candidate effect modifiers (age, parental education, family income) that were not stable

across iterations and warrant replication in independent cohorts before they are read as confirmed. The reproducible signal was a binary sex difference in early-period suicidal-behavior risk at SSRI initiation. This identifies sex as a candidate axis of caution to weigh in the risk-benefit discussion at SSRI initiation, beyond the age-based stratification of the FDA’s existing pediatric black-box warning.

Declaration of AI use

I used Claude Code and OpenAI ChatGPT to assist in:

- Converting the original SAS data extraction scripts into R,
- Implementing the data extraction and analysis pipelines,
- Checking for bugs in the implementation,
- Interpreting the results and generating ideas on how to improve them,
- Checking for errors in the written report.

References

- [1] Zachary Steel, Claire Marnane, Changiz Iranpour, Tien Chey, John W. Jackson, Vikram Patel, and Derrick Silove. The global prevalence of common mental disorders: a systematic review and meta-analysis 1980–2013. *International Journal of Epidemiology*, 43(2):476–493, 2014. doi: 10.1093/ije/dyu038.
- [2] Cheng-Hao Yang, Jia-Jie Lv, Xiang-Meng Kong, Feng Chu, Zhi-Bin Li, Wei Lu, and Xin-Yu Li. Global, regional and national burdens of depression in adolescents and young adults aged 10–24 years, from 1990 to 2019: findings from the 2019 Global Burden of Disease study. *British Journal of Psychiatry*, 225(2):311–320, 2024. doi: 10.1192/bjp.2024.69.
- [3] National Institute for Health and Care Excellence. Depression in children and young people: identification and management. NICE guideline [NG134], 2019. URL <https://www.nice.org.uk/guidance/ng134>.
- [4] Tyra Lagerberg, Yasmina Molero, Brian M. D’Onofrio, Lina Politi de la Cruz, Paul Lichtenstein, Lorena Fernandez de la Cruz, Clara Hellner, and Zheng Chang. Antidepressant prescription patterns and CNS polypharmacy with antidepressants among children, adolescents, and young adults: a population-based study in Sweden. *European Child & Adolescent Psychiatry*, 28(8):1137–1145, 2019. doi: 10.1007/s00787-018-01269-2.
- [5] Lotte Rasmussen, Peter B. Jensen, Johan Reutfors, Kari Furu, Svetlana Skurtveit, Randi Selmer, Per Damkier, Mette Bliddal, and Rikke Wesselhoeft. Treatment patterns of antidepressants in children and adolescents in Scandinavia. *European Child & Adolescent Psychiatry*, 34(1):159–167, 2024. doi: 10.1007/s00787-024-02433-7.
- [6] Tarek A. Hammad, Thomas Laughren, and Judith Racoosin. Suicidality in pediatric patients treated with antidepressant drugs. *Archives of General Psychiatry*, 63(3):332–339, 2006. doi: 10.1001/archpsyc.63.3.332.
- [7] Philippe Courtet and Jorge Lopez-Castroman. Antidepressants and suicide risk in depression. *World Psychiatry*, 16(3):317–318, 2017. doi: 10.1002/wps.20460.
- [8] Jeffrey A. Bridge, Satish Iyengar, Cheryl B. Salary, Remy P. Barbe, Boris Birmaher, Cynthia R. Pfeffer, and David A. Brent. Clinical response and risk for reported suicidal ideation and suicide attempts in pediatric antidepressant treatment: a meta-analysis of randomized controlled trials. *JAMA*, 297(15):1683–1696, 2007. doi: 10.1001/jama.297.15.1683.
- [9] Tyra Lagerberg, Anthony A. Matthews, Nanbo Zhu, Seena Fazel, Juan J. Carrero, and Zheng Chang. Effect of selective serotonin reuptake inhibitor treatment following

- diagnosis of depression on suicidal behaviour risk: a target trial emulation. *Neuropsychopharmacology*, 48(12):1760–1768, 2023. doi: 10.1038/s41386-023-01676-3.
- [10] U.S. Food and Drug Administration. Prozac (fluoxetine hydrochloride) capsules, for oral use — highlights of prescribing information. Drugs@FDA, NDA 020031, supplement 082, 2023. URL https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/020031s0821b1.pdf. Boxed warning on suicidal thoughts and behaviors in patients aged 24 years and younger.
 - [11] Jakob A. Kaminski and Tom Bschor. Antidepressants and suicidality: A re-analysis of the re-analysis. *Journal of Affective Disorders*, 266:95–99, 2020. doi: 10.1016/j.jad.2020.01.107.
 - [12] Michael A. Posternak, Mark Zimmerman, Gabor I. Keitner, and Ivan W. Miller. A reevaluation of the exclusion criteria used in antidepressant efficacy trials. *American Journal of Psychiatry*, 159(2):191–200, 2002. doi: 10.1176/appi.ajp.159.2.191.
 - [13] U.S. Food and Drug Administration. Major depressive disorder: Developing drugs for treatment. Draft Guidance for Industry, 2018. URL <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/major-depressive-disorder-developing-drugs-treatment>.
 - [14] Anthony A. Matthews, Goodarz Danaei, Nazrul Islam, and Tobias Kurth. Target trial emulation: applying principles of randomised trials to observational studies. *BMJ*, 378:e071108, 2022. doi: 10.1136/bmj-2022-071108.
 - [15] Miguel A. Hernán and James M. Robins. Using big data to emulate a target trial when a randomized trial is not available. *American Journal of Epidemiology*, 183(8): 758–764, 2016. doi: 10.1093/aje/kwv254.
 - [16] Barbra A. Dickerman, Xabier García-Albéniz, Roger W. Logan, Spiros Denaxas, and Miguel A. Hernán. Avoidable flaws in observational analyses: an application to statins and cancer. *Nature Medicine*, 25(10):1601–1606, 2019. doi: 10.1038/s41591-019-0597-x.
 - [17] Stefan Feuerriegel, Dennis Frauen, Valentyn Melnychuk, Jonas Schweisthal, Konstantin Hess, Alicia Curth, Stefan Bauer, Niki Kilbertus, Isaac S. Kohane, and Mhaela van der Schaar. Causal machine learning for predicting treatment outcomes. *Nature Medicine*, 30(4):958–968, 2024. doi: 10.1038/s41591-024-02902-1.
 - [18] David M. Kent, Jessica K. Paulus, David van Klaveren, Ralph D’Agostino, Steve Goodman, Rodney Hayward, John P. A. Ioannidis, Bray Patrick-Lake, Sally Morton, Michael Pencina, Gowri Raman, Joseph S. Ross, Harry P. Selker, Ravi Varadhan, Andrew Vickers, John B. Wong, and Ewout W. Steyerberg. The predictive

- approaches to treatment effect heterogeneity (PATH) statement. *Annals of Internal Medicine*, 172(1):35–45, 2020. doi: 10.7326/M18-3667.
- [19] Sara T. Brookes, Elise Whitely, Matthias Egger, George Davey Smith, Paul A. Mulheran, and Tim J. Peters. Subgroup analyses in randomized trials: risks of subgroup-specific analyses; power and sample size for the interaction test. *Journal of Clinical Epidemiology*, 57(3):229–236, 2004. doi: 10.1016/j.jclinepi.2003.08.009.
 - [20] Susan Athey, Julie Tibshirani, and Stefan Wager. Generalized random forests. *The Annals of Statistics*, 47(2):1148–1178, 2019. doi: 10.1214/18-AOS1709.
 - [21] Stefan Wager and Susan Athey. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242, 2018. doi: 10.1080/01621459.2017.1319839.
 - [22] Erik Sverdrup, Maria Petukhova, and Stefan Wager. Estimating treatment effect heterogeneity in psychiatry: A review and tutorial with causal forests. *International Journal of Methods in Psychiatric Research*, 34(2):e70015, 2025. doi: 10.1002/mpr.70015.
 - [23] Tiansheng Wang, Alexander P. Keil, Siyeon Kim, Richard Wyss, Phyo Than Htoo, Michele Jonsson Funk, John B. Buse, Michael R. Kosorok, and Til Stürmer. Iterative causal forest: A novel algorithm for subgroup identification. *American Journal of Epidemiology*, 193(5):764–776, 2024. doi: 10.1093/aje/kwad219.
 - [24] Tiansheng Wang, Virginia Pate, Richard Wyss, John B. Buse, Michael R. Kosorok, and Til Stürmer. High-dimensional iterative causal forest (hdicf) for subgroup identification using health care claims data. *American Journal of Epidemiology*, 194(7):2085–2097, 2025. doi: 10.1093/aje/kwae322.
 - [25] J. Michael Bostwick and V. Shane Pankratz. Affective disorders and suicide risk: A reexamination. *American Journal of Psychiatry*, 157(12):1925–1932, 2000. doi: 10.1176/appi.ajp.157.12.1925.
 - [26] Xinkun Nie and Stefan Wager. Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika*, 108(2):299–319, 2021. doi: 10.1093/biomet/asaa076.
 - [27] Leo Breiman, Jerome H. Friedman, Richard A. Olshen, and Charles J. Stone. *Classification and Regression Trees*. Wadsworth, Belmont, CA, 1984. See §3.4.3 for the 1-SE rule used to select tree complexity when cross-validation error does not uniquely identify a model.
 - [28] Fabian Termorshuizen, Saskia J.M. Palmen, and Eibert R. Heerdink. Suicide behavior before and after the start with antidepressants: A high persistent risk in the first

- month of treatment among the young. *International Journal of Neuropsychopharmacology*, 19(2):pyv081, 2016. doi: 10.1093/ijnp/pyv081.
- [29] Arif Khan, Amy E. Brodhead, Kelly A. Schwartz, Russell L. Kolts, and Walter A. Brown. Sex differences in antidepressant response in recent antidepressant clinical trials. *Journal of Clinical Psychopharmacology*, 25(4):318–324, 2005. doi: 10.1097/01.jcp.0000168879.03169.ce.
- [30] Rolf H. H. Groenwold, Ian R. White, A. Rogier T. Donders, James R. Carpenter, Douglas G. Altman, and Karel G. M. Moons. Missing covariate data in clinical research: when and when not to use the missing-indicator method for analysis. *CMAJ*, 184(11):1265–1269, 2012. doi: 10.1503/cmaj.110977.

A Supplementary tables

A.1 Variable name mapping

Table 13 maps the raw variable names used in the analysis code (e.g., R script outputs, log files, intermediate RDS files) to the human-readable labels used in the body of this report.

Table 13: Mapping between raw variable names and human-readable labels

Raw name	Human-readable label
<i>Treatment and outcome</i>	
cc	SSRI initiation indicator
sb12_itt	Suicidal behavior outcome (12-week ITT)
fu_start	Index date / time zero / baseline
fu_end_itt	End of ITT follow-up (event, death, emigration, or week 12)
diagn_date	Date of first recorded depression diagnosis
<i>Demographics and SES</i>	
female	Sex
age	Age
age_cat	Age category
year	Calendar year
source	Care setting
edufam_cat	Parental education
inc_cat	Family income
<i>Family history</i>	
fh_suicidal	Family history of suicidal behavior (count of biological parents)
fh_depr	Family history of depression (count of biological parents)
<i>Prior psychiatric diagnoses</i>	
diag_organic	Organic mental disorder
diag_alcohol	Alcohol use disorder
diag_sud	Substance use disorder
diag_psychotic	Psychotic disorder
diag_bipolar	Bipolar disorder
diag_mdd	Major depressive disorder (not used as covariate; constant=1)

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(Table 13 continued)

Raw name	Human-readable label
diag_phobic	Phobic anxiety disorder
diag_anxiety_other	Other anxiety disorder
diag_anxiety	Combined F40+F41 anxiety indicator
diag OCD	Obsessive-compulsive disorder
diag_stress	Stress/adjustment disorder
diag_anorexia	Anorexia nervosa
diag_bulimia	Bulimia nervosa
diag_sleep	Sleep disorder
diag_personality_cluster_b	Cluster B personality disorder
diag_intellectual_disability	Intellectual disability
diag_autism	Autism spectrum disorder
diag_adhd	ADHD
diag_conduct	Conduct disorder
diag_overdose	Overdose/poisoning
diag_suicidal	Prior suicidal behavior
<i>Prior medications (within 90 days before baseline)</i>	
med_antipsychotic	Antipsychotic
med_hypnotic	Hypnotic/sedative
med_benzodiazepine	Benzodiazepine
med_antiepileptic	Antiepileptic
med_mood_stabilizer	Mood stabilizer
med_stimulant	Psychostimulant
med_opioid	Opioid
med_addiction	Addiction medication
<i>Healthcare utilization</i>	
hosp	Prior psychiatric hospitalization
<i>HD features (hdiCF only)</i>	
dx.inp.<code>	Inpatient ICD-10 3-character code (e.g., dx.inp.F32)
dx.out.<code>	Outpatient ICD-10 3-character code (e.g., dx.out.F40)
rx.<code>	ATC 5-character chemical-subgroup code dispensed within 90 days (e.g., rx.N05BA)

A.2 Missingness patterns in the four CCA covariates

The complete-case analysis cohort (Section 2.6) excludes patients with any missing value in the four partially-observed covariates: parental education, family income, family history of suicidal behavior, and family history of depression. Across the 93,795 patients in the pre-CCA eligible cohort, only 5 of the $2^4 = 16$ possible missingness patterns are observed (Table 14). The remaining 11 patterns are empty.

Table 14: Missingness patterns across the four partially-observed covariates that drive the complete-case analysis cohort. PE = parental education; FI = family income; FH-S = family history of suicidal behavior; FH-D = family history of depression. ✓ = observed; × = missing. Percentages are out of the pre-CCA eligible cohort (N = 93,795). Patterns with zero patients are omitted.

Pattern	PE	FI	FH-S	FH-D	N	%
None missing	✓	✓	✓	✓	88,427	94.28
Family income missing	✓	×	✓	✓	953	1.02
Parental education missing	×	✓	✓	✓	218	0.23
Parental education + family income missing	×	×	✓	✓	274	0.29
All four missing	×	×	×	×	3,923	4.18

A.3 Baseline characteristics of patients excluded by the complete-case restriction

Table 15 compares the 5,368 patients dropped by the complete-case restriction against the 88,427 patients retained in the analysis cohort, on demographics, the index-diagnosis care setting, the four partially-observed covariates, and two outcome-adjacent prior diagnoses. The SMD column quantifies the magnitude of any systematic difference between the excluded and the included groups on each variable.

Table 15: Baseline characteristics of the 5,368 patients excluded by the complete-case restriction (Excluded column) compared to the 88,427 patients retained in the analysis cohort (Included column). SMD = absolute standardized mean difference between the two groups. Values >0.1 indicate notable imbalance.

Characteristic	Excluded (N = 5,368)	Included (N = 88,427)	SMD
<i>Demographics</i>			
Female, n (%)	2,679 (49.9)	54,730 (61.9)	0.24

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(Table 15 continued)

Characteristic	Excluded (N = 5,368)	Included (N = 88,427)	SMD
Age category (years)			0.29
6–11	73 (1.4)	2,288 (2.6)	
12–17	1,730 (32.2)	39,919 (45.1)	
18–24	3,565 (66.4)	46,220 (52.3)	
<i>Source of depression diagnosis</i>			
Care setting			0.19
Outpatient	4,492 (83.7)	79,612 (90.0)	
Inpatient	821 (15.3)	8,440 (9.5)	
Other/unknown source	55 (1.0)	375 (0.4)	
<i>Partially-observed covariates</i>			
Parental education			3.20
Primary	244 (4.5)	4,842 (5.5)	
Secondary	408 (7.6)	40,678 (46.0)	
Post-secondary	301 (5.6)	42,907 (48.5)	
Missing	4,415 (82.2)	0 (0.0)	
Family income			7.16
Negative	2 (0.0)	167 (0.2)	
Zero	15 (0.3)	137 (0.2)	
1st–20th percentile	115 (2.1)	16,625 (18.8)	
20th–80th percentile	81 (1.5)	53,650 (60.7)	
>80th percentile	5 (0.1)	17,848 (20.2)	
Missing	5,150 (95.9)	0 (0.0)	
Family history of suicidal behavior			2.33
Neither parent	1,351 (25.2)	79,633 (90.1)	
One parent	89 (1.7)	8,347 (9.4)	
Both parents	5 (0.1)	447 (0.5)	
Missing	3,923 (73.1)	0 (0.0)	
Family history of depression			2.36
Neither parent	1,243 (23.2)	65,675 (74.3)	
One parent	194 (3.6)	20,576 (23.3)	
Both parents	8 (0.1)	2,176 (2.5)	
Missing	3,923 (73.1)	0 (0.0)	
<i>Outcome-adjacent prior diagnoses</i>			
Prior suicidal behavior, n (%)	555 (10.3)	7,539 (8.5)	0.06
Overdose/poisoning, n (%)	432 (8.0)	5,927 (6.7)	0.05

A.4 High-dimensional features for hdiCF

Table 16 lists the 39 automatically generated high-dimensional features used by the hdiCF pipeline, grouped by source and ordered by population prevalence within each group. Inpatient and outpatient diagnoses are 3-character ICD-10 codes recorded on or before the index date. Prescription features are 5-character (chemical-subgroup) ATC codes dispensed in the 90 days up to and including the index date, with antidepressants (N06A) excluded by construction (Section 2.8). Codes with prevalence below 1% were filtered out during feature generation.

Table 16: Generated high-dimensional features used by hdiCF, with population prevalence (%).

Feature	Description	Prevalence (%)
<i>Inpatient ICD-10</i>		
dx.inp.F32	Depressive episode	11.96
dx.inp.F10	Alcohol use disorder	3.53
dx.inp.X64	Intentional self-poisoning by drugs (NOS)	3.20
dx.inp.T50	Poisoning by drugs and biological substances	3.04
dx.inp.F43	Reaction to severe stress / adjustment disorder	2.90
dx.inp.F90	Hyperkinetic disorder (ADHD)	2.25
dx.inp.F19	Multiple substance use disorder	1.50
dx.inp.F50	Eating disorder	1.36
dx.inp.F84	Autism spectrum disorder	1.00
<i>Outpatient ICD-10</i>		
dx.out.F32	Depressive episode	84.88
dx.out.F90	Hyperkinetic disorder (ADHD)	13.44
dx.out.F33	Recurrent depressive disorder	9.18
dx.out.F43	Reaction to severe stress / adjustment disorder	8.63
dx.out.F50	Eating disorder	6.94
dx.out.F41	Other anxiety disorder	5.71
dx.out.F84	Autism spectrum disorder	5.26
dx.out.F40	Phobic anxiety disorder	4.37
dx.out.F98	Other behavioural / emotional disorder of childhood	3.75
dx.out.F51	Non-organic sleep disorder	3.66
dx.out.F10	Alcohol use disorder	3.59
dx.out.F42	Obsessive-compulsive disorder	2.56

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(Table 16 continued)

Feature	Description	Prevalence (%)
dx.out.F12	Cannabinoid use disorder	1.89
dx.out.F19	Multiple substance use disorder	1.84
dx.out.F91	Conduct disorder	1.77
dx.out.T50	Poisoning by drugs and biological substances	1.55
dx.out.F60	Specific personality disorder	1.52
dx.out.X64	Intentional self-poisoning by drugs (NOS)	1.51
dx.out.X78	Intentional self-harm by sharp object	1.11
dx.out.F31	Bipolar affective disorder	1.08
dx.out.F70	Mild intellectual disability	1.05
<i>ATC (90 days before baseline)</i>		
rx.N05B	Anxiolytics	16.73
rx.N05CH	Melatonin receptor agonists	10.00
rx.N05C	Hypnotics and sedatives	6.41
rx.N06BA	Centrally acting sympathomimetics (ADHD stimulants)	6.10
rx.N05CF	Benzodiazepine-related hypnotics (Z-drugs)	5.74
rx.N05BA	Benzodiazepine anxiolytics	3.03
rx.N05AH	Atypical antipsychotics (diazepines / thiazepines)	1.94
rx.N03AX	Other antiepileptics	1.21
rx.N05AX	Other antipsychotics	1.08

A.5 Auto-tuner instability of the raw causal forest

Section 2.8 reports that the raw causal forest used for variable selection is grown with `tune.parameters = "none"` because the `grf` auto-tuner picks essentially at random across RNG seeds on this cohort. Table 17 provides the underlying evidence: five raw-CF fits to the same data ($N = 88,427$, 36 candidate covariates), using the same code path as the production pipeline but with `tune.parameters = "all"` and seeds $\{42, 43, 44, 45, 46\}$, each producing its own tuner-selected configuration and downstream above-mean variable-importance selection set.

Table 17: Hyperparameters chosen by `grf`'s auto-tuner across five RNG seeds for the raw causal forest used in iCF variable selection. Each column is one fit to the same data under `tune.parameters = "all"`. The bottom row is the size of the above-mean variable-importance selection set that would be passed to the iCF iterative stage. The selection set ranges from 4 to 14 variables, all at essentially identical CV fit (relative range of CV-MSE across seeds $\sim 10^{-4}$).

Hyperparameter	Seed 42	Seed 43	Seed 44	Seed 45	Seed 46
<code>sample.fraction</code>	0.462	0.433	0.300	0.325	0.061
<code>mtry</code>	17	21	19	14	10
<code>min.node.size</code>	70	38	561	106	472
<code>honesty.fraction</code>	0.557	0.686	0.649	0.727	0.686
<code>honesty.prune.leaves</code>	0	1	1	1	0
<code>alpha</code>	0.014	0.099	0.018	0.008	0.172
<code>imbalance.penalty</code>	1.119	1.472	0.405	0.837	0.292
Above-mean selection-set size	13	9	12	14	4

The most extreme draw (seed 46, `sample.fraction` = 0.06) is an aggressive-shrinkage configuration in which 32 of the 36 candidate covariates received zero variable importance. In contrast, seed 45 retained 14 variables above the importance mean. Disabling the auto-tuner and fixing `seed = 42L` eliminates this source of nondeterminism: two consecutive runs of the production pipeline produce byte-identical variable importance.

B Supplementary figures

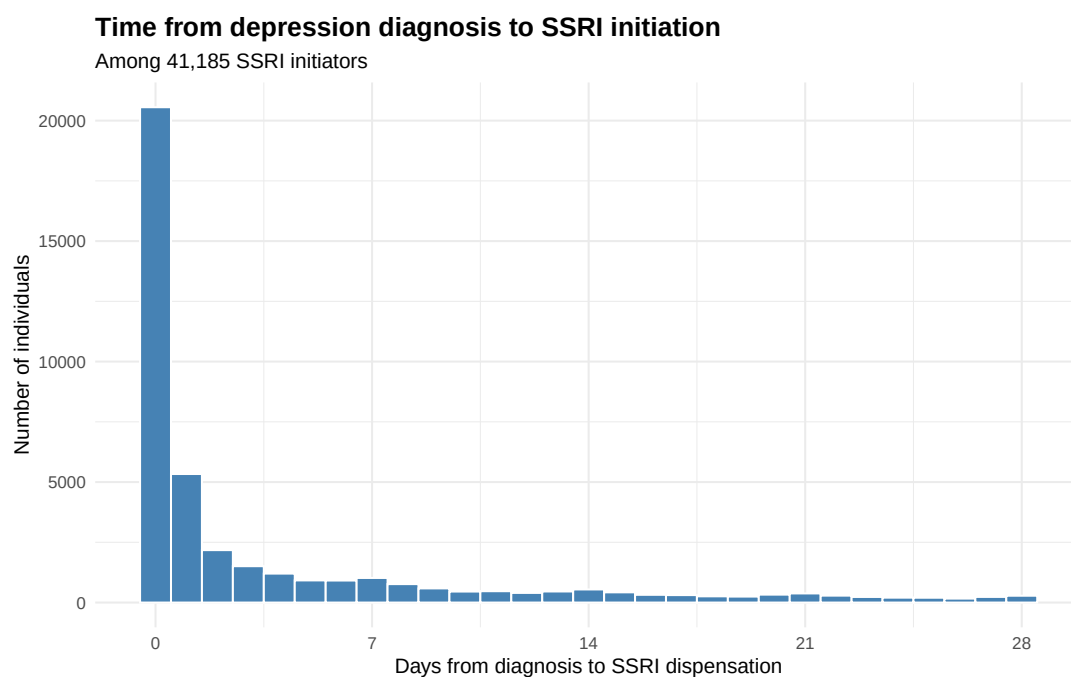


Figure 30: Distribution of time from depression diagnosis to SSRI dispensation among the 41,185 initiators in the complete-case analysis cohort (same denominator as Table 2). Approximately half initiated SSRI on the same day as their diagnosis (day 0). The 28-day grace period captures the vast majority of initiations, with a steep drop-off after the first week.